

## SUPPLEMENTARY MATERIAL

# Supplementary material for: GoCo: structure-guided conformal calibration for error-controlled Gene Ontology annotation release

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## Abstract

This supplement contains the proofs supporting the main-text risk guarantee and a definition of the GoDag baseline (Supplementary Note S1), the construction and implementation details of the method—split mechanics, interpretation of the frozen GO reference, the per-call ranking model, the arithmetic behind Remark S2, the uniform grid path and a worked micro-example of one block (Supplementary Note S2)—and the second structural instantiation with its validation without retuning and ranking-fold selection rule (Supplementary Note S3). Tables report split-level operating characteristics (Table S1); the error-target sweep at  $\delta = 0.10$  (Table S2); certification, pool and held-out risks with yield diagnostics (Table S3); block diagnostics (Table S4); illustrative released genes (Table S5); the gene-level versus pooled-call estimand comparison (Table S6); source-object structures (Table S7); gains over the matched-risk frontier in the neighbourhood-structure family (Table S8); the ranking-fold selection rule (Table S9); commonly used release rules on all six predictors (Table S10); prespecified constants (Table S11); the method inventory (Table S12); and the  $\delta = 0.50$  error-target sweep (Table S13). Figure S1 gives the primary comparison at  $\delta = 0.10$ . The full construction and additional formal statements are given in Supplementary Note S4, with AI-use disclosure in Supplementary Note S5. Main-text cross-references are explicitly identified; equations and formal results labelled S refer to this supplement.

**Key words:** Gene Ontology; conformal risk control; Learn-Then-Test

## Supplementary Note S1: Proofs

### S1.1 Setting and notation

The gene panel  $\mathcal{U} = \{1, \dots, n\}$ , the score matrix  $(s_{ig})$ , the GO snapshot and the reference annotations are fixed. For every release policy  $\pi$  and gene  $i$  the loss  $L_i(\pi) = L_i(S_i(\pi)) \in [0, 1]$  of Equation (1) is therefore a fixed number. The only random object is a uniformly distributed permutation  $\varsigma$  of  $\mathcal{U}$ . With  $n_{\mathcal{T}} = \text{round}(0.10n)$  and  $n_{\mathcal{T}} + m = \text{round}(0.70n)$ , the ranking fold is  $\mathcal{T} = \varsigma(\{1, \dots, n_{\mathcal{T}}\})$ , the certification fold is  $\mathcal{C} = \varsigma(\{n_{\mathcal{T}} + 1, \dots, n_{\mathcal{T}} + m\})$  and the evaluation fold is  $\mathcal{E} = \varsigma(\{n_{\mathcal{T}} + m + 1, \dots, n\})$ . The pool is  $\mathcal{P} = \mathcal{U} \setminus \mathcal{T}$  with  $N = |\mathcal{P}| = m + n_{\mathcal{E}}$  and  $f = m/N$ . For a policy  $\pi$  write  $R(\pi) = N^{-1} \sum_{i \in \mathcal{P}} L_i(\pi)$ ,  $\hat{R}_{\mathcal{C}}(\pi) = m^{-1} \sum_{i \in \mathcal{C}} L_i(\pi)$ ,

$\hat{R}_{\mathcal{E}}(\pi) = n_{\mathcal{E}}^{-1} \sum_{i \in \mathcal{E}} L_i(\pi)$ ,  $\sigma^2(\pi) = N^{-1} \sum_{i \in \mathcal{P}} \{L_i(\pi) - R(\pi)\}^2$  and  $S^2(\pi) = N\sigma^2(\pi)/(N - 1)$ . For a candidate  $\pi_k$  abbreviate  $\hat{R}_k = \hat{R}_{\mathcal{C}}(\pi_k)$  and  $\hat{\sigma}_k^2 = m^{-1} \sum_{i \in \mathcal{C}} \{L_i(\pi_k) - \hat{R}_k\}^2$  (divisor  $m$ , matching `ddof=0` in the implementation), and let  $\bar{\sigma}_k = \max(\hat{\sigma}_k, 10^{-6})$  and  $z_{\delta} = \Phi^{-1}(\delta)$ .

### S1.2 The split as a sampling design

**Lemma S1** (Conditional sampling design) *Conditional on  $\mathcal{T}$ , the certification fold  $\mathcal{C}$  is uniformly distributed over the  $m$ -element subsets of  $\mathcal{P}$ , i.e. a simple random sample without replacement (SRSWOR) of size  $m$  from  $\mathcal{P}$ , and  $\mathcal{E} = \mathcal{P} \setminus \mathcal{C}$  is a SRSWOR of size  $n_{\mathcal{E}}$  from  $\mathcal{P}$ . Unconditionally,  $\mathcal{T} \cup \mathcal{C}$  is a SRSWOR of size  $n_{\mathcal{T}} + m$  from  $\mathcal{U}$ .*

*Proof* Fix  $T_0$  with  $|T_0| = n_{\mathcal{T}}$  and an  $m$ -subset  $C_0 \subseteq \mathcal{U} \setminus T_0$ . The number of permutations with  $\varsigma(\{1, \dots, n_{\mathcal{T}}\}) = T_0$  and  $\varsigma(\{n_{\mathcal{T}} + 1, \dots, n_{\mathcal{T}} + m\}) = C_0$  is  $n_{\mathcal{T}}! m! n_{\mathcal{E}}!$ , independent of  $C_0$ ; hence all  $\binom{N}{m}$  subsets are equally likely given  $\mathcal{T} = T_0$ . The complement of a uniform  $m$ -subset of  $\mathcal{P}$  is a uniform  $n_{\mathcal{E}}$ -subset. The last claim follows in the same way from the count  $(n_{\mathcal{T}} + m)! n_{\mathcal{E}}!$ .  $\square$

**Lemma S2** (Moments under SRSWOR) *Let  $b_i \in [0, 1]$ ,  $i \in \mathcal{P}$ , be fixed numbers with mean  $\bar{b}_{\mathcal{P}}$  and  $S_b^2 = (N-1)^{-1} \sum_{i \in \mathcal{P}} (b_i - \bar{b}_{\mathcal{P}})^2$ . If  $\mathcal{C}$  is a SRSWOR of size  $m$  from  $\mathcal{P}$  and  $\bar{b}_{\mathcal{C}} = m^{-1} \sum_{i \in \mathcal{C}} b_i$ , then  $\mathbb{E} \bar{b}_{\mathcal{C}} = \bar{b}_{\mathcal{P}}$ ,  $\text{Var} \bar{b}_{\mathcal{C}} = (1-f) S_b^2 / m \leq 1/(2m)$ , and with  $\bar{b}_{\mathcal{E}}$  the mean over  $\mathcal{E}$ ,  $N \bar{b}_{\mathcal{P}} = m \bar{b}_{\mathcal{C}} + n_{\mathcal{E}} \bar{b}_{\mathcal{E}}$ .*

*Proof* Let  $I_i = \mathbf{1}\{i \in \mathcal{C}\}$ ; then  $\Pr(I_i = 1) = f$  and  $\Pr(I_i = I_j = 1) = m(m-1)/\{N(N-1)\}$  for  $i \neq j$ , so  $\text{Var} I_i = f(1-f)$  and  $\text{Cov}(I_i, I_j) = -f(1-f)/(N-1)$ . Unbiasedness follows from  $\mathbb{E} \sum_i I_i b_i = f \sum_i b_i$ . For the variance,  $\text{Var}(\sum_i I_i b_i) = f(1-f) \{\sum_i b_i^2 - (N-1)^{-1} \sum_{i \neq j} b_i b_j\} = f(1-f) N S_b^2$ , and dividing by  $m^2 = f^2 N^2$  gives  $(1-f) S_b^2 / m$ . Since  $b_i \in [0, 1]$ ,  $\sigma_b^2 \leq 1/4$  and  $S_b^2 \leq N/[4(N-1)] \leq 1/2$  for  $N \geq 2$ ; the variance formula is Theorem 2.2 of Cochran (1977). The identity is  $\sum_{\mathcal{P}} b_i = \sum_{\mathcal{C}} b_i + \sum_{\mathcal{E}} b_i$ .  $\square$

### S1.3 Finite-population central limit theorem

**Lemma S3** (Hájek 1960; Erdős and Rényi 1959) *For each  $t$  let  $b_{t,i} \in [0, 1]$ ,  $i \in \mathcal{P}_t$ ,  $|\mathcal{P}_t| = N_t$ , be fixed numbers with mean  $\bar{b}_t$ , variance  $\sigma_t^2 = N_t^{-1} \sum_i (b_{t,i} - \bar{b}_t)^2$  and  $S_t^2 = N_t \sigma_t^2 / (N_t - 1)$ , and let  $\mathcal{C}_t$  be a SRSWOR of size  $m_t$  from  $\mathcal{P}_t$  with  $f_t = m_t / N_t$ . If  $m_t(1-f_t)\sigma_t^2 \rightarrow \infty$ , then*

$$W_t = \frac{\bar{b}_{\mathcal{C}_t} - \bar{b}_t}{\sqrt{(1-f_t)S_t^2/m_t}} \xrightarrow{d} \mathcal{N}(0, 1).$$

*Proof* Let  $s_t^2 = \text{Var} \sum_{i \in \mathcal{C}_t} b_{t,i} = m_t(1-f_t)S_t^2$  (Lemma S2). Hájek (1960) proved that  $W_t$  is asymptotically standard normal if and only if, for every  $\tau > 0$ ,  $\{\sum_i (b_{t,i} - \bar{b}_t)^2\}^{-1} \sum_{i: |b_{t,i} - \bar{b}_t| > \tau s_t} (b_{t,i} - \bar{b}_t)^2 \rightarrow 0$ ; Erdős and Rényi (1959) gave a sufficient condition of the same type. Since  $|b_{t,i} - \bar{b}_t| \leq 1$  and  $s_t^2 \geq m_t(1-f_t)\sigma_t^2 \rightarrow \infty$ , the index set  $\{i : |b_{t,i} - \bar{b}_t| > \tau s_t\}$  is empty once  $\tau s_t > 1$ , so the condition holds. Note that  $m_t(1-f_t)\sigma_t^2 \rightarrow \infty$  forces  $m_t \rightarrow \infty$  and  $N_t - m_t \rightarrow \infty$ , because  $\sigma_t^2 \leq 1/4$  and  $m_t(1-f_t) \leq \min(m_t, N_t - m_t)$ .  $\square$

### S1.4 Structure of the candidate path

**Lemma S4** (Nestedness and  $\mathcal{T}$ -measurability) *Let  $B_j$ ,  $\mathcal{T}_j$ ,  $u_i$ ,  $h_i$ ,  $k(q)$ ,  $Q_{j,q}$  and  $\pi_{j,q}$  be as in Supplementary Note S4.1. (i) For every block  $j$ , every  $q \leq q'$  in  $\mathcal{Q} \cup \{1\}$  and every gene  $i$ ,  $S_i(\lambda_j) \subseteq S_i(\pi_{j,q}) \subseteq S_i(\pi_{j,q'}) \subseteq S_i(\lambda_{j+1})$ ; consequently the GoCo path satisfies  $S_i(\pi_k) \subseteq S_i(\pi_{k+1})$  for all  $i, k$ . (ii) The sequence  $\pi_1, \dots, \pi_K$ , including  $K$ , is a function of  $\mathcal{T}$  and frozen objects only, and  $K \leq J + (J-1)|\mathcal{Q}|$ .*

*Proof* (i) Because  $\lambda_j > \lambda_{j+1}$ ,  $S_i^H = S_i(\lambda_j) \subseteq S_i(\lambda_{j+1}) = S_i^L$ . For  $i \notin B_j$  all four sets coincide. For  $i \in B_j$ ,  $S_i(\pi_{j,q}) \in \{S_i^H, S_i^L\}$  with  $S_i(\pi_{j,q}) = S_i^L$  exactly when  $i \in Q_{j,q}$ , so it suffices to show  $Q_{j,q} \subseteq Q_{j,q'}$ . The lexicographic order on  $(u_i, h_i, \text{id}(i))$  is a deterministic total order on  $B_j$ . For  $q \leq q' < 1$ ,  $k(q) = \max\{1, |q| \mathcal{T}_j|\} \leq k(q')$ , so the  $k(q)$ -th smallest ranking-fold triple precedes or equals the  $k(q')$ -th,

whence  $Q_{j,q} \subseteq Q_{j,q'}$ . If  $q' = 1$ , then  $Q_{j,1} = B_j$  by definition, so the inclusion also holds. Consecutive elements of the path are  $(\lambda_j, \pi_{j,q_{\min}})$ ,  $(\pi_{j,q}, \pi_{j,q'})$  with  $q < q' < 1$ ,  $(\pi_{j,q_{\max}}, \pi_{j,1})$  or  $(\lambda_j, \lambda_{j+1})$ ; because  $\pi_{j,1} = \lambda_{j+1}$ , the chain of inclusions gives the claim.

(ii) The grid and the sets  $S_i^H, S_i^L$ , and hence  $c_i$ ,  $D_i$ ,  $B_j$  and the source-attribution weights  $a_{ir}$ , are functions of frozen predictor objects. The true quantities  $\Delta_i$  and  $\phi_i$  are label-dependent; in constructing the path they are evaluated only for ranking-fold genes  $i \in \mathcal{T}_j$ . The block means and the source estimates of Equation (S3) therefore use only ranking-fold labels. The per-call model behind Equation (S4) is fitted on ranking-fold calls alone, and its features are frozen predictor output together with rate estimates built from ranking-fold labels, so the fitted  $\hat{p}_{ig}$  are  $\mathcal{T}$ -measurable; hence so are  $\hat{\Delta}_i$  and  $\hat{\tau}_i$  for a pool gene, which sum those probabilities over that gene's frozen added calls  $D_i$ , and so is  $u_i = \hat{\Delta}_i / \max(\hat{\tau}_i, 10^{-3})$ ;  $h_i$  is a fixed function of the gene identifier. For  $q < 1$ , the cutoff  $(u^*, h^*, \text{id}^*)$  is an order statistic among ranking-fold genes and  $Q_{j,q}$  compares each affected gene's fixed-given- $\mathcal{T}$  triple with it; the endpoint  $Q_{j,1} = B_j$  is fixed by definition. The insertion triggers compare  $|\mathcal{T}|^{-1} \sum_{i \in \mathcal{T}} L_i(\lambda_{j+1})$  with  $0.8\alpha$  and  $|\mathcal{T}_j|$  with 5. Thus every decision shaping the sequence is determined by  $\mathcal{T}$ ; each of the  $J$  thresholds is visited once and each of the  $J-1$  blocks contributes at most  $|\mathcal{Q}|$  policies. If  $\mathcal{T}_j$  is empty the trigger fails and no adaptive policy is inserted, so Equation (S3) is never evaluated on an empty fold.  $\square$

Nestedness of the released sets does not imply monotonicity of the losses:  $L_i(\pi_{k+1}) < L_i(\pi_k)$  is possible when added terms are supported. Only the total number of released calls is monotone along the path; the validity argument below uses neither property.

### S1.5 Consistency of the variance estimator

**Lemma S5** (Consistency of  $\hat{\sigma}_k$ ) *Condition on  $\mathcal{T}$  and fix a policy  $\pi$  with pool values  $b_i = L_i(\pi)$ , pool variance  $\sigma^2(\pi)$  and sample variance  $\hat{\sigma}^2 = m^{-1} \sum_{i \in \mathcal{C}} b_i^2 - \bar{b}_{\mathcal{C}}^2$ . If  $m \rightarrow \infty$  then  $\hat{\sigma}^2 - \sigma^2(\pi) \rightarrow 0$  in probability. If in addition  $\liminf \sigma^2(\pi) > 0$ , then  $\hat{\sigma}/S(\pi) \rightarrow 1$  in probability.*

*Proof* Apply Lemma S2 to  $b_i$  and to  $b_i^2$ , both in  $[0, 1]$ : each sample mean has variance at most  $1/(2m) \rightarrow 0$ , so by Chebyshev's inequality  $\bar{b}_{\mathcal{C}} \rightarrow \bar{b}_{\mathcal{P}}$  and  $m^{-1} \sum_{\mathcal{C}} b_i^2 \rightarrow N^{-1} \sum_{\mathcal{P}} b_i^2$  in probability; hence  $\hat{\sigma}^2 \rightarrow \sigma^2(\pi)$ . If  $\liminf \sigma^2(\pi) > 0$ , the continuous-mapping theorem gives  $\hat{\sigma}/\sigma(\pi) \rightarrow 1$ , and since  $S(\pi)/\sigma(\pi) = \sqrt{N/(N-1)} \rightarrow 1$ , it follows that  $\hat{\sigma}/S(\pi) \rightarrow 1$  in probability.  $\square$

### S1.6 Proof of main-text Proposition 1

*Proof* Condition on  $\mathcal{T}$  throughout. By Lemma S4(ii) the sequence  $\pi_1, \dots, \pi_K$  is fixed and by Lemma S1  $\mathcal{C}$  is a SRSWOR of size  $m$  from  $\mathcal{P}$ . The rule returns  $\hat{\pi} = \pi_{\hat{k}}$  with  $\hat{k} = \max\{k : p_1 \leq \delta, \dots, p_k \leq \delta\}$ , or abstains (risk  $0 \leq \alpha$ ) if  $p_1 > \delta$ .

*Step 1: reduction to the first invalid candidate.* Let  $\mathcal{K}_0 = \{k : R(\pi_k) > \alpha\}$ . If  $\mathcal{K}_0 = \emptyset$  there is nothing to prove. Otherwise let  $k^* = \min \mathcal{K}_0$ , a function of  $\mathcal{T}$ . On  $\{R(\hat{\pi}) > \alpha\}$  a policy was certified and  $\hat{k} \in \mathcal{K}_0$ , so  $\hat{k} \geq k^*$  and, by definition of  $\hat{k}$ ,  $p_k \leq \delta$  for all  $k \leq \hat{k}$ , in particular  $p_{k^*} \leq \delta$ . Hence  $\Pr\{R(\hat{\pi}) > \alpha \mid \mathcal{T}\} \leq \Pr\{p_{k^*} \leq \delta \mid \mathcal{T}\}$ . This step uses only the stopping rule; the joint distribution of  $(p_1, \dots, p_K)$ , all computed from the same fold, plays no role, and neither does nestedness.

*Step 2: asymptotic level at  $k^*$ .* Write  $k = k^*$ ,  $b_i = L_i(\pi_k)$  and  $S_k^2 = S^2(\pi_k)$ . Since  $\Phi$  is increasing,  $p_k \leq \delta$  if and only if

$$\sqrt{m}(\hat{R}_k - \alpha) \leq z_\delta \tilde{\sigma}_k.$$

Because  $R(\pi_k) > \alpha$ , on the event  $\{p_k \leq \delta\}$ ,

$$\sqrt{m}\{\hat{R}_k - R(\pi_k)\} < \sqrt{m}(\hat{R}_k - \alpha) \leq z_\delta \tilde{\sigma}_k.$$

Since  $\delta \leq \frac{1}{2}$ ,  $z_\delta \leq 0$ , and since  $\tilde{\sigma}_k = \max(\hat{\sigma}_k, 10^{-6}) \geq \hat{\sigma}_k$ ,

$$z_\delta \tilde{\sigma}_k \leq z_\delta \hat{\sigma}_k.$$

Thus, on  $\{\hat{\sigma}_k > 0\}$ ,

$$\{p_k \leq \delta\} \subseteq \left\{ \frac{\sqrt{m}\{\hat{R}_k - R(\pi_k)\}}{\hat{\sigma}_k} \leq z_\delta \right\}.$$

The hypotheses  $m \rightarrow \infty$ ,  $N - m \rightarrow \infty$  and  $\liminf S_k^2 > 0$  imply  $m(1 - f_t)\sigma_k^2 \rightarrow \infty$ , where  $f_t = m/N$ , so Lemma S3 gives

$$W_k = \frac{\hat{R}_k - R(\pi_k)}{\sqrt{(1 - f_t)S_k^2/m}} \xrightarrow{d} \mathcal{N}(0, 1).$$

By Lemma S5,  $\hat{\sigma}_k/S_k \rightarrow 1$  in probability, and therefore, by Slutsky's theorem,

$$\frac{1}{\sqrt{1 - f_t}} \frac{\sqrt{m}\{\hat{R}_k - R(\pi_k)\}}{\hat{\sigma}_k} = W_k \frac{S_k}{\hat{\sigma}_k} \xrightarrow{d} \mathcal{N}(0, 1).$$

Moreover, Lemma S5 and the variance assumption imply  $\Pr(\hat{\sigma}_k = 0 \mid \mathcal{T}) \rightarrow 0$ . Hence, with  $f_t \rightarrow f$ ,

$$\limsup \Pr(p_k \leq \delta \mid \mathcal{T}) \leq \Phi\left(\frac{z_\delta}{\sqrt{1 - f}}\right).$$

For  $\delta \leq \frac{1}{2}$ ,  $z_\delta \leq 0$  and  $\sqrt{1 - f} \leq 1$ , so  $\Phi(z_\delta/\sqrt{1 - f}) \leq \Phi(z_\delta) = \delta$ , with equality iff  $z_\delta = 0$  ( $\delta = \frac{1}{2}$ ) or  $f = 0$ . Combining with Step 1 proves the conditional bound. The  $o(1)$  is in  $m$ : it collects the normal-approximation error for  $W_k$  and the Slutsky remainder for a single,  $\mathcal{T}$ -determined index; the path length  $K$  enters nowhere.

*Numerical value.* When the variance floor is asymptotically inactive,  $\hat{\sigma}_k/S_k \rightarrow 1$ , so the ratio of the scale used by Equation (S7) to the without-replacement standard error approaches  $(1 - f)^{-1/2} \approx 1.73$  at  $f = 2/3$ , making the normal test conservative. At  $\delta = 0.10$ , main-text Proposition 1 therefore gives the asymptotic upper bound  $\Phi(-1.2816/\sqrt{1/3}) = \Phi(-2.22) \approx 0.0132$ ; at  $\delta = 0.50$  the formal upper bound is 0.50. These are upper bounds, not claims about the actual failure probability of the selected procedure.

*Step 3: ordering-free validity (Corollary S1).* Steps 1–2 used three properties of the candidate sequence: it is fixed given  $\mathcal{T}$ ;  $K$  is finite; and the pool variance is bounded below at the first invalid candidate. The first holds for any rule that orders affected genes by a statistic computed from  $\mathcal{T}$ , frozen objects and randomisation independent of the split (hash only; mean added-call score;  $\hat{\Delta}_i/c_i^c$ ; Equation (S4)), with or without the insertion triggers, which are themselves  $\mathcal{T}$ -measurable. Nothing in the proof refers to the association between the ordering and  $\Delta_i$ , which therefore affects only which index  $\hat{k}$  is reached.

*Step 4: unconditional statement.* Steps 1–2 give  $\Pr\{R(\hat{\pi}) > \alpha \mid \mathcal{T}\} \leq \Phi(z_\delta/\sqrt{1 - f}) + \varepsilon_m(\mathcal{T})$  with  $\varepsilon_m(\mathcal{T}) \rightarrow 0$ . If the pool variance at the first invalid candidate is bounded below

uniformly in  $\mathcal{T}$ , a Berry–Esseen bound for sampling without replacement (Höglund 1978), whose remainder depends on the population only through moment ratios that are bounded for values in  $[0, 1]$  with variance bounded below, makes  $\varepsilon_m$  uniform in  $\mathcal{T}$ , and  $\Pr\{R(\hat{\pi}) > \alpha\} = \mathbb{E}[\Pr\{R(\hat{\pi}) > \alpha \mid \mathcal{T}\}] \leq \Phi(z_\delta/\sqrt{1 - f}) + o(1)$ .

*Global methods.* In the module-family analyses, for GoDag and the global baselines that use Equation (S7), the candidate sequence is the deterministic grid, the calibration sample  $\mathcal{T} \cup \mathcal{C}$  is a SRSWOR of size  $n_{\mathcal{T}} + m$  from  $\mathcal{U}$  (Lemma S1) and the estimand is the panel mean  $n^{-1} \sum_{i \in \mathcal{U}} L_i(\pi)$ ; Steps 1–2 apply verbatim with  $\mathcal{P}$  replaced by  $\mathcal{U}$  and  $f$  by 0.7. In the neighbourhood-family analyses, the deterministic global paths are certified on  $\mathcal{C}$ , so Steps 1–2 apply directly with pool  $\mathcal{P}$  and  $f = m/N$ . Network-HAC variance substitution is not covered by this argument for  $\delta < \frac{1}{2}$ ; at  $\delta = \frac{1}{2}$  a normal-type rule with positive scale reduces to the same empirical-mean check. Empirical Bernstein and Hoeffding-type bounds use different tests, so this cancellation does not apply to them.

*Finite-sample alternative.* Hoeffding's inequality holds for the mean of a SRSWOR of bounded values (Hoeffding 1963, Theorem 4), so  $p_k^H = \exp\{-2m(\alpha - \hat{R}_k)_+^2\}$  is exactly super-uniform under  $H_k$  conditional on  $\mathcal{T}$ , and Step 1 gives a finite-sample bound  $\delta$  for the same path at the cost of a wider margin ( $\sqrt{\log(1/\delta)/(2m)} \approx 0.011$  at  $\delta = 0.10$ ,  $m = 9,477$ , versus 0.0038 for Equation (S7)).  $\square$

## S1.7 Proof of Remark S2

*Proof  $\delta$  identity.* Since  $\Phi$  is strictly increasing and  $\tilde{\sigma}_k > 0$ ,  $p_k \leq \delta \iff \sqrt{m}(\hat{R}_k - \alpha)/\tilde{\sigma}_k \leq z_\delta \iff \hat{R}_k \leq \alpha + z_\delta \tilde{\sigma}_k/\sqrt{m}$ . At  $\delta = 0.50$ ,  $z_\delta = 0$ . At  $\delta = 0.10$ ,  $z_\delta = -1.2816$ ; with  $\tilde{\sigma}_k = 0.29$  and  $m = 9,477$ ,  $|z_\delta|\tilde{\sigma}_k/\sqrt{m} = 0.0038$ . Since  $\hat{R}_k$  depends only on  $\mathcal{C}$  and the candidate, any two methods that share a path, a loss and a certification fold and differ only in a normal-type variance estimator certify the same policies at  $\delta = 0.50$ .

*Split identity.* By Lemma S2 with  $b_i = L_i(\pi)$ ,  $NR(\pi) = m\hat{R}_C(\pi) + n_{\mathcal{E}}\hat{R}_{\mathcal{E}}(\pi)$  for every policy, including  $\pi = \hat{\pi}$  (the identity is algebraic and holds whether or not the policy was selected using  $\mathcal{C}$ ). Subtracting  $N\alpha = m\alpha + n_{\mathcal{E}}\alpha$  and dividing by  $n_{\mathcal{E}}$  gives  $\hat{R}_{\mathcal{E}}(\pi) - \alpha = (N/n_{\mathcal{E}})\{R(\pi) - \alpha\} - (m/n_{\mathcal{E}})\{\hat{R}_C(\pi) - \alpha\}$ . In the 10/60/30 design  $N/n_{\mathcal{E}} = 3.000$  and  $m/n_{\mathcal{E}} = 2.000$  up to rounding (Wainberg: 14,215/4,738 and 9,477/4,738); these are also the factors for the neighbourhood-family global methods, which certify on  $\mathcal{C}$ . For module-family global methods, with  $n$  in place of  $N$  and  $n_{\mathcal{T}} + m$  in place of  $m$ , the factors are 10/3 and 7/3.

*Exceedance at  $\delta = 0.50$ .* If a policy is certified at  $\delta = 0.50$  then  $\hat{R}_C(\hat{\pi}) \leq \alpha$ , so the second term is nonnegative and  $\hat{R}_{\mathcal{E}}(\hat{\pi}) - \alpha \geq (N/n_{\mathcal{E}})\{R(\hat{\pi}) - \alpha\}$ ; hence  $\{R(\hat{\pi}) > \alpha\} \subseteq \{\hat{R}_{\mathcal{E}}(\hat{\pi}) > \alpha\}$  and the held-out exceedance frequency bounds the pool-level failure frequency from above. The converse fails:  $\hat{R}_{\mathcal{E}}(\hat{\pi}) > \alpha$  also occurs when  $R(\hat{\pi}) \leq \alpha$  but  $\hat{R}_C(\hat{\pi})$  is sufficiently far below  $\alpha$ . Under SRSWOR the standard deviation of  $\hat{R}_C(\pi)$  for a fixed policy with  $\sigma(\pi) = 0.29$  is  $\sqrt{(1 - f)/m} \sigma \approx 0.0017$  in Wainberg; because the fixed-sequence rule stops at the first candidate whose certification mean exceeds  $\alpha$ , the selected candidate typically has  $\hat{R}_C(\hat{\pi})$  a few thousandths below  $\alpha$  (a selection effect),  $R(\hat{\pi})$  within a few thousandths of  $\alpha$ , and the evaluation-fold mean above  $\alpha$  in a substantial fraction of splits.

*Exceedance at  $\delta = 0.10$ .* More generally, certification gives  $\hat{R}_C(\hat{\pi}) - \alpha \leq z_\delta \tilde{\sigma}/\sqrt{m}$ , and the exact split identity therefore gives

$$\hat{R}_{\mathcal{E}}(\hat{\pi}) - \alpha \geq \frac{N}{n_{\mathcal{E}}} \{R(\hat{\pi}) - \alpha\} - \frac{m}{n_{\mathcal{E}}} \frac{z_\delta \tilde{\sigma}}{\sqrt{m}}.$$

For illustration, with  $\bar{\sigma} = 0.29$  and  $m = 9,477$ , the certification margin is approximately 0.0038, giving the corresponding approximations 0.0076 and 0.0025 under the Wainberg fold ratios. These numerical margins are illustrative and candidate-specific. A held-out exceedance does not, in general, imply  $R(\hat{\pi}) > \alpha$ ; conversely, for a certified policy with  $\delta \leq 0.50$ ,  $R(\hat{\pi}) > \alpha$  implies  $\hat{R}_E(\hat{\pi}) > \alpha$ .  $\square$

#### S1.8 Proof of Lemma S6

*Proof* Condition on  $\mathcal{T}$ , fix a block and write  $B = B_j \cap \mathcal{P}$ ,  $M = |B|$ .

(a) *Granularity.* For a policy that relaxes exactly the set  $Q \subseteq B$ ,  $L_i(\pi) = L_i(S_i^H) + \Delta_i$  for  $i \in Q$  and  $L_i(\pi) = L_i(S_i^H)$  otherwise, so  $R(\pi) = R(S^H) + N^{-1} \sum_{i \in Q} \Delta_i$  by linearity of the pool mean; with  $Q = \{i : \text{rk}(i) \leq k\}$  this is the first claim. If  $Q^{\text{rand}}$  is a uniformly random  $k$ -subset of  $B$ , each  $i \in B$  belongs to it with probability  $k/M$ , so  $\mathbb{E} \sum_{i \in Q^{\text{rand}}} \Delta_i = k \bar{\Delta}_B = (k/M) \sum_{i \in B} \Delta_i$ , and  $\sum_{i \in B} \Delta_i = N \{R(S^L) - R(S^H)\}$  because genes outside  $B$  have  $\Delta_i = 0$ . For the global threshold,  $S_i(\lambda) = S_i^H \cup \{g \in D_i : s_{ig} \geq \lambda\}$  for  $\lambda \in [\lambda_{j+1}, \lambda_j]$ , so the released sets change only when  $\lambda$  crosses one of the  $r$  distinct scores in the block, giving at most  $r - 1$  policies strictly between the endpoints; if all added rows share one score  $s^*$ , then  $S_i(\lambda) = S_i^H$  for every  $i \in B$  when  $\lambda > s^*$  and  $S_i(\lambda) = S_i^L$  for every  $i \in B$  when  $\lambda \leq s^*$ , so no global threshold separates affected genes. In Wainberg 2,470 of the 2,484 rows of the block  $800 \rightarrow 775$  share the score 789.65.

(b) *Ordering value.* With  $Q(k) = \{i : \text{rk}(i) \leq k\}$ ,  $\sum_{i \in Q(k)} \Delta_i - \mathbb{E} \sum_{i \in Q^{\text{rand}}} \Delta_i = \sum_{i \in B} (\Delta_i - \bar{\Delta}_B) \mathbf{1}\{\text{rk}(i) \leq k\} = M \text{Cov}_B \{\mathbf{1}\{\text{rk}(i) \leq k\}, \Delta_i\}$ , where  $\text{Cov}_B(x, y) = M^{-1} \sum_{i \in B} (x_i - \bar{x}_B)(y_i - \bar{y}_B) = M^{-1} \sum_{i \in B} (x_i - \bar{x}_B)y_i$ ; dividing by  $N$  gives the first identity. Summing over  $k = 1, \dots, M$  and using  $\sum_{k=1}^M \mathbf{1}\{\text{rk}(i) \leq k\} = M + 1 - \text{rk}(i)$ ,

$$\begin{aligned} & \frac{1}{M} \sum_{k=1}^M \left\{ \sum_{i \in Q(k)} \Delta_i - \mathbb{E} \sum_{i \in Q^{\text{rand}}} \Delta_i \right\} \\ &= \frac{1}{M} \sum_{i \in B} (\Delta_i - \bar{\Delta}_B) \{M + 1 - \text{rk}(i)\} \\ &= -\text{Cov}_B \{\text{rk}(i), \Delta_i\}, \end{aligned}$$

so the average risk saving of the ordering is  $N^{-1} \text{Cov}_B \{\text{rk}(i), \Delta_i\}$ , positive exactly when late-admitted genes tend to have larger increments. For a uniformly random ordering independent of the increments,  $\mathbb{E} \text{rk}(i) = (M+1)/2$  for every  $i$ , so the expected covariance is zero. If  $\Delta_i$  is nondecreasing in  $\text{rk}(i)$ , then for every  $k$  the first  $k$  genes have the  $k$  smallest increments, so  $\sum_{i \in Q(k)} \Delta_i$  is minimal among all  $k$ -subsets and in particular at most its random expectation (Chebyshev's sum inequality). Ties at the cutoff are resolved by the fixed gene hash to provide a deterministic reproducible total order. The random-order expectation above is a theoretical benchmark and does not require or imply that this fixed hash is probabilistically independent of  $\Delta$ .  $\square$

**Remark S1** (Interpretation and empirical counterparts) *Lemma S6 is a statement about fixed numbers given  $\mathcal{T}$ . It explains how partial admission and an informative ordering can reduce fixed-pool risk at a given admission level, but it does not guarantee a larger certified prefix or an optimal yield because certification also depends on sampling variation and first-failure stopping. Part (a) describes granularity that a global threshold can forgo, especially within an exact tie,*

*and part (b) expresses the ordering contribution through a rank-increment covariance on unseen genes. Whether that covariance is favourable for the ranking-fold-trained  $u_i$  on pool genes is an empirical question; the out-of-sample Spearman correlation between  $u_i$  and  $\Delta_i$  over affected pool genes is reported per certified block in Table S4. GoCo-M and GoCo-N are two prespecified candidate-path constructions whose relative performance is assessed empirically.*

#### S1.9 The GoDag baseline

GoDag is the hierarchy-aware global-threshold comparator used throughout: its candidate path is a deterministic score grid and its loss is the gene-level FDP of Equation (1) with the TruePath acceptable set. In the module-family analyses it calibrates on the first 70% of each permutation, whereas in the neighbourhood-family analyses it certifies on  $\mathcal{C}$  (60%) to match the other methods. The GoDag rows of the main text use the normal  $p$ -value of Equation (S7); replacing it by the corresponding Hoeffding bound with the appropriate calibration-sample size gives an exact finite-sample certificate at a cost in yield.

#### S1.10 References for this note

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## Supplementary Note S2: Construction and implementation details

This note collects the construction and implementation details of the main text: the split mechanics (Section S2.1), the interpretation of the frozen TruePath reference (Section S2.2), the source-level estimate and the per-call ranking model (Section S2.3), the derivation arithmetic behind Remark S2 (Section S2.4), the uniform grid path and the choice of ranking statistic (Section S2.5), the worked micro-example of a block (Section S2.6), further conventions and definitions (Section S2.7), source-object generalization (Section S2.8), and the gene-level case study of the released annotations (Section S2.9). Equations and results are cross-referenced to their current locations.

### S2.1 Splits: seeds, rounding and realised fold sizes

Each of the 100 analysis streams draws a uniformly random permutation of the panel (seed 20260905 +  $s$  for stream  $s$  in the module family; the neighbourhood family of Supplementary Note S3 uses seed 20260910 +  $s$  with the same cut) and cuts it into a *ranking fold*  $\mathcal{T}$  (the first  $n_{\mathcal{T}} = \text{round}(0.10n)$  genes; its labels build the ordering and candidate path), a *certification fold*  $\mathcal{C}$  (the next  $m = \text{round}(0.70n) - n_{\mathcal{T}}$  genes, about 60%) and an *evaluation fold*  $\mathcal{E}$  (the remaining  $n_{\mathcal{E}}$  genes, about 30%; rounding is the implementation's round-half-to-even). The realised sizes ( $n_{\mathcal{T}}, m, n_{\mathcal{E}}$ ) are (1,579, 9,477, 4,738) in Wainberg, (1,488, 8,932, 4,465) in Sanger, (628, 3,770, 1,885) in DRIVE and (1,502, 9,012, 4,506) in HAP1. The pool of non-ranking genes is  $\mathcal{P} = \mathcal{C} \cup \mathcal{E}$  with  $N = n - n_{\mathcal{T}}$  and  $f = m/N \approx 2/3$ . GoCo uses  $\mathcal{T}$  only to build its candidate path and  $\mathcal{C}$  only to certify it. In the module family, global methods including GoDag have no ranking step and calibrate on  $\mathcal{T} \cup \mathcal{C}$  (70%); in the neighbourhood family, all compared methods certify on the same  $\mathcal{C}$  (60%) fold. Evaluation-fold labels are used by no method before evaluation; the upstream predictor is frozen and treated as a constant.

### S2.2 Interpretation of the frozen TruePath reference

The main analysis uses the frozen TruePath reference throughout. A released term is counted as *unsupported* when it is absent from that reference for the gene after TruePath closure. This is an operational evaluation label, not a claim that the biological function is absent. Because GO is open-world and incomplete, the biological interpretation of reference-unsupported calls is treated separately in the Discussion; all methods here are evaluated against the same frozen reference.

### S2.3 The per-call ranking model

*Where the estimate comes from.* The structure that creates the ties also supplies the evidence for the estimate. A call  $(i, g)$  exists because the annotated members  $\mathcal{R}_{ig}$  of gene  $i$ 's winning module carry term  $g$ ; genes sharing module evidence can be offered overlapping terms by overlapping sets of members. If those members generated unsupported terms for ranking-fold genes, they are likely to do so again for  $i$ . Let  $\hat{p}_r$  be the fraction of ranking-fold calls supported by member  $r$  whose term is unsupported, shrunk toward the overall ranking-fold rate as in Equation (S3) with the weight of two pseudo-calls, and let  $\hat{p}_{ig}$  be the mean of  $\hat{p}_r$  over  $r \in \mathcal{R}_{ig}$ . This is the co-essentiality signal GoCo uses; Equation (S3) could equally be applied to gene-level increments through the attribution matrix  $a_{ir}$ ; any such variant is  $\mathcal{T}$ -measurable and so covered by Corollary S1.

*Features.* GoCo fits a logistic regression of the indicator “call unsupported” on seven per-call features, over all ranking-fold calls at or above the lowest grid threshold. Three are the co-essentiality signal and its companions, learned from ranking-fold labels only:  $\hat{p}_{ig}$ ; the shrunk unsupported rate of the term  $g$  itself across ranking-fold calls; and the log frequency of  $g$  among the TruePath annotations of ranking-fold genes, a proxy for how specific the term is. Four are frozen predictor output: the log score, the number of supporting members  $|\mathcal{R}_{ig}|$ , the number of calls the gene has above the grid floor, and the module size excluding the target (Wainberg) or the target-module correlation (Sanger, DRIVE) or the cluster  $z$ -score (HAP1). We deliberately exclude the predictor's background count: in Wainberg it is the term's annotation count *excluding the target*, so it would encode the target's own label.

*Cross-fitting and hyperparameters.* For ranking-fold calls, the classifier fits and the raw term- and source-rate counts are cross-fitted between two hash-defined halves. The module implementation uses a common overall unsupported-rate shrinkage target computed from the whole ranking fold for both halves, and the term-frequency feature is computed from whole-ranking-fold TruePath counts with the scored gene removed. Thus the ranking-fold predictions are cross-fitted at the classifier and raw-rate-count levels but are not fully held-half-exclusive for every label-derived summary. Pool calls are scored by the model fitted on the whole ranking fold. Writing  $\hat{p}_{ig}$  for the fitted probabilities, define  $\hat{L}_i(S) = |S|^{-1} \sum_{g \in S} \hat{p}_{ig}$  for  $|S| > 0$  and  $\hat{L}_i(\emptyset) = 0$ . The estimates entering Equation (S4) are  $\hat{\Delta}_i = \hat{L}_i(S_i^L) - \hat{L}_i(S_i^H)$  and  $\hat{\tau}_i = \sum_{g \in \mathcal{D}_i} (1 - \hat{p}_{ig})$ . The feature set, the model class and its hyperparameters ( $C = 0.5$  on standardised features) were fixed on ten Wainberg development splits (streams 0–9) before the external analyses; the neighbourhood-family instantiation was fixed on FunMap streams 0–9. The model-class check replaces the logistic model by a histogram gradient-boosting classifier (150 trees, depth 3, learning rate 0.06). All label-derived quantities remain functions of  $\mathcal{T}$  alone: no certification- or evaluation-fold labels enter the ordering, so Corollary S1 applies.

### S2.4 Remark S2: the split identity and held-out exceedance

The split identity gives, for every policy,

$$\hat{R}_{\mathcal{E}}(\pi) - \alpha = \frac{N}{n_{\mathcal{E}}} \{R(\pi) - \alpha\} - \frac{m}{n_{\mathcal{E}}} \{\hat{R}_{\mathcal{C}}(\pi) - \alpha\},$$

with  $(N/n_{\mathcal{E}}, m/n_{\mathcal{E}}) = (3, 2)$  for GoCo and for all methods in the neighbourhood-family matched-certification analysis, and  $(10/3, 7/3)$  for module-family global methods, up to rounding. At  $\delta = 0.50$  the certified policy is the last one before the first candidate whose certification-fold mean exceeds  $\alpha$ ; on a fine path  $\hat{R}_{\mathcal{C}}(\hat{\pi})$  sits just below  $\alpha$ ,  $R(\hat{\pi})$  lies within about one standard error ( $\hat{\sigma}\sqrt{(1-f)/m} \approx 0.0017$ ) of  $\alpha$ , and the evaluation-fold mean overshoots  $R(\hat{\pi})$  by twice the certification-fold undershoot. Held-out exceedance of  $\alpha$  in a substantial fraction of GoCo splits at  $\delta = 0.50$  is therefore expected—19–42% across the four data sets (Table S1)—and it is an upper bound on the pool-level failure frequency: if  $\hat{R}_{\mathcal{C}}(\hat{\pi}) \leq \alpha$  then  $R(\hat{\pi}) > \alpha$  implies  $\hat{R}_{\mathcal{E}}(\hat{\pi}) > \alpha$ . Because every label in this benchmark is known,  $R(\hat{\pi})$  is computable in each split, and the fraction of splits with  $R(\hat{\pi}) > \alpha$  is the direct empirical check of main-text Proposition 1. GoDag on the Wainberg grid stops at 800 with a calibration mean  $\hat{R}_{\mathcal{T} \cup \mathcal{C}} \approx 0.092$ , far below  $\alpha$ , which



explains both its 1% exceedance and its unused budget; on the finer external grids GoDag’s held-out exceedance (36% and 28% in Sanger and DRIVE) is comparable to GoCo’s. Section S1.7 proves these statements.

## S2.5 The uniform grid path and the choice of ranking statistic

Two design choices define the construction certified here. First, the candidate path is defined uniformly: the same grid as GoDag with partial admission inserted in every step that meets a ranking-fold trigger, rather than a hand-picked block. This makes later thresholds available after inserted policies and helps explain the observed gains, but does not guarantee a larger certified prefix: a construction restricted to a single block could not move beyond it when a larger target permits further release. Second, the ordering statistic is the predicted loss increment per predicted supported term, used as a cost-per-gain heuristic for prioritising candidates expected to add less gene-level FDP per supported term; its form and the model behind it were fixed on the development data and applied unchanged externally.

## S2.6 A worked micro-example of one block

The Wainberg grid step from  $\lambda_j = 800$  to  $\lambda_{j+1} = 775$  is the block that GoCo splits at  $\alpha = 0.10$ . It contains 2,484 gene–GO rows, of which 2,470 share the score 789.650 exactly and the remaining 14 are spread over the scores 797.626, 783.124 and 779.901; the affected set  $B_j$  contains 1,267 genes, of which about 127 fall in the ranking fold. The dominant 2,470-call exact tie at score 789.650 cannot be divided by any global threshold; the remaining 14 rows provide only three additional score locations within the step. The full-panel TruePath risk moves from 0.0924 at threshold 800 to 0.1398 at 775. GoCo instead evaluates, for each admission fraction  $q$  in the grid  $\mathcal{Q}$ , the policy that gives  $S_i^L$  to the  $q$ -quantile of affected genes with the smallest  $u_i$  of Equation (S4) and  $S_i^H$  to the rest. In the full-panel linear interpolation of Lemma S6(a), the risk reaches  $\alpha = 0.10$  at about  $q = 0.16$ ; this is not itself a certification calculation. The GoCo ordering bends the path below that interpolation and was certified at  $q = 0.35$  or  $q = 0.40$  in 89 of the 100 splits ( $q = 0.40$  in 64 of them). In split 0 the certified policy admitted 40% of the affected genes.

*Where the ties come from, in detail.* The enrichment score of a Wainberg call is a function of three small integers: the number  $k_{\text{sup}}$  of annotated module members supporting the term, the module size  $n_{\text{mod}}$  excluding the target, and the background count  $K_{\text{bg}}$ . Because 85% of scored calls come from modules with  $n_{\text{mod}} \leq 5$ , many (module, term) pairs share a count triple and therefore a score. The 2,470 calls at 789.65 belong to 1,256 genes, 482 terms and 347 different modules, and 2,018 of them have  $(k_{\text{sup}}, n_{\text{mod}}, K_{\text{bg}}) = (1, 4, 5)$ . The external resources tie for the same reason: Sanger has 1,880 pairs at one value among the 29,120 pairs at or above score 200, spread over 847 genes and 393 modules; DRIVE 723 of 9,258; HAP1 14,245 of 122,262. The Sanger and DRIVE grid steps used at  $\alpha = 0.10$  contain 20 and 36 distinct values, so a finer global grid would recover part of the lost granularity there, whereas the HAP1 step  $475 \rightarrow 450$  is again almost a single tie: 2,089 of its 2,115 calls share one value. Full multiplicities are in `tie_counts.csv` in the reproducibility package.

## S2.7 Further conventions and definitions

*What the estimand does and does not control.* The certified risk  $R(\pi)$  averages the per-gene FDP of Equation (1) over all

pool genes, with abstaining genes contributing zero. The loss penalizes reference-unsupported terms among those released but does not penalize reference-supported terms that are withheld; reference-supported yield therefore measures retained supported functions separately from the error rate; it is not a false-negative rate. In the module-structure analyses, only about 8–12% of genes receive any term at  $\alpha = 0.10$ , so the average is dominated by abstentions and constrains no individual annotated gene: because abstention carries no loss, a policy that releases entirely unsupported sets for 9% of pool genes and abstains on the rest has  $R = 0.09$  and is admissible at  $\alpha = 0.10$ . Nor does  $R(\pi)$  bound the pooled fraction of unsupported gene–GO calls, which weights genes by set size. GoCo does not control this pooled quantity. Table S6 includes a pair-based conformal-selection comparator targeting pooled call-level FDR as a descriptive comparison; because our random split samples genes rather than individual gene–GO pairs, the usual pair-level exchangeability conditions are not established here and we make no formal pooled-call FDR guarantee for these benchmark runs. We certify  $R(\pi)$  and report, for every method, the number of released calls, the number of annotated genes, the reference-supported term count and the supported terms per supported gene as descriptive yield diagnostics (Table S3).

*The three qualifications of main-text Proposition 1.* The result is asymptotic in  $m$  ( $m = 3,770$ – $9,477$  here); the path length is irrelevant because a single test matters. It controls the pool risk  $R(\hat{\pi})$ , not the evaluation-fold mean, which is itself a noisy estimate of  $R(\hat{\pi})$  (Remark S2). At  $\delta = 0.50$  the formal bound is  $\frac{1}{2}$  and the normal certificate reduces to the empirical calibration check; we use this setting as the primary benchmark, while  $\delta = 0.10$  provides the stricter certification analysis.

*The target-excluded enrichment background.* One property of the Wainberg predictor deserves mention: its enrichment background excludes the target gene, so a gene that is itself annotated with a term receives a slightly larger background-corrected score for that term (by a factor  $K/(K - 1)$  for a term with  $K$  annotated panel genes). This is a feature of the published predictor, enters the frozen scores of every method identically, and is not used by any ordering beyond the score itself; we excluded the raw background count from the learned features for this reason. For GoCo-M, the grid, insertion triggers, admission grid,  $\kappa$ , ordering family and learned model specification were fixed on Wainberg; the neighbourhood-family instantiation and its model specification were fixed on FunMap streams 0–9. These choices are listed in Table S11 and none affects validity.

*Score comparison and reporting.* Scores are compared after rounding to six decimals. Besides the means and exceedance frequencies of the main text we report, for each paired comparison, the fraction of splits in which one method exceeds another.

*Source sets.* The sources entering Equation (S2) are restricted to the gene panel, and a gene whose added calls had no retained source would fall back to the block mean of Equation (S3). In all four data sets every added call has at least one retained source, so  $\sum_r a_{ir} = 1$  for every affected gene in every certified block and this fallback is never used (Table S4).

*Sign of the increment.* A gene moving out of abstention has  $S_i^H = \emptyset$  and  $\Delta_i$  equal to the false discovery proportion of its added calls, which is nonnegative;  $\Delta_i < 0$  occurs only when  $S_i^H$  is nonempty and the added calls are better supported than the existing set.

*Compatibility of the gene-averaged target.* Because abstention costs nothing, a gene-averaged target of 0.10 is compatible with a low supported fraction of the released call list, which is why the released-call and supported-term counts are reported beside the certified risk.

*What the ordering has to estimate.* Admitting gene  $i$  changes pool risk by  $\Delta_i/N$  and gains  $\tau_i = |D_i \cap A_i^{\text{TP}}|$  reference-supported terms. We use  $\Delta_i/\tau_i$  as a cost-per-gain ordering heuristic, prioritising genes expected to add less gene-level FDP per supported term; this ordering is not claimed to solve the discrete allocation problem exactly. Neither quantity can be evaluated for a gene the certificate will later use because both depend on that gene's labels. The ordering therefore estimates this ratio from the ranking fold and frozen predictor output, and Corollary S1 ensures that the quality of that estimate affects yield rather than validity.

*Evaluation and reporting.* For every method and split we record, on the evaluation fold, the mean gene-level TruePath FDP (a held-out assessment of the selected policy), the number of genes with a nonempty set, the number of released calls, the supported fraction of calls, the supported-gene yield (evaluation-fold genes with at least one TruePath-supported released term) and the GO-term yield (supported evaluation-fold gene–GO calls); on the certification fold, the certification mean; and the pool risk  $R(\hat{\pi})$  of the certified policy, which every label in the benchmark makes computable and which main-text Proposition 1 controls asymptotically. The pool risk is the mean loss over the calibration sample together with its corresponding evaluation fold: the whole panel for module-family global methods, and  $\mathcal{C} \cup \mathcal{E}$  for GoCo and for all methods in the neighbourhood-family matched-certification comparison. For descriptive comparison the tables report this quantity using the TruePath loss for every method. Multilabel certifies the Direct loss instead, so for it the event formally controlled by its certificate is the Direct-loss pool risk exceeding  $\alpha$ ; those frequencies are reported separately in the footnote of Table S1. Because the Direct acceptable set is contained in its TruePath closure, the Direct loss is at least as large as the TruePath loss for the same release. Reported are means over the 100 splits, the fraction of splits with  $\hat{R}_{\mathcal{E}} > \alpha$  (held-out exceedance) and with  $R(\hat{\pi}) > \alpha$  (pool-level failure, the event bounded by main-text Proposition 1), and paired split-wise differences in GO-term yield with the fraction of splits in which one method exceeds another. Splits are overlapping random partitions of one fixed panel, and every reported quantity concerns comparisons of release rules on that panel rather than a biological superpopulation. Variation across the 100 split streams reflects the randomisation design and Monte Carlo error; an exceedance or failure frequency  $f$  estimated from 100 splits has Monte Carlo standard error  $\sqrt{f(1-f)/100} \leq 0.05$ , so frequencies of 0.00–0.01 are consistent with, but cannot resolve, the asymptotic boundary values of main-text Proposition 1. The most liberal candidate on every path is the predictor's own reporting floor—score 600 in Wainberg, 100 in the three external resources and  $-\log_{10} q = 1$  in FunMap and STRING—and it is worth recording when that floor binds, since a comparison in which the global baseline runs out of grid is not a comparison of methods. Over the 100 splits it binds only in DRIVE at  $\alpha = 0.20$ , where at  $\delta = 0.50$  every method reaches the end of the grid in every split (which is why the difference there is exactly zero; at  $\delta = 0.10$  GoCo-M stops one step short in one split), and in one Sanger split at  $\alpha = 0.20$ ,  $\delta = 0.50$ . Elsewhere the certified cutoff sits well above it: 625–650 in Wainberg at  $\alpha = 0.20$ , and 16–62 against a floor of

1 in FunMap and STRING, so extending either grid further down—as a coarser published grid would—changes no arm and no comparison.

*Comparators.* Every method uses the same 100 split streams, frozen scores and held-out TruePath truth within each data set. *Multilabel* is our re-implementation of the released multilabel statistic of Boger et al. (one-sided normal  $p$ -value with empirical standard deviation,  $\delta = 0.50$  default) on the Direct loss; it calibrates on  $\mathcal{T} \cup \mathcal{C}$  in the module family and on  $\mathcal{C}$  in the neighbourhood family, and ignores the ontology. *GoDag* applies the same certificate to the TruePath loss and selects a single global threshold, using the same family-specific calibration sample. The generic Direct-loss calibrators are reported in the corresponding main-text comparison tables. *GoCo-M* adds predictor-native shared-evidence structure through the candidate path of Supplementary Note S4.1, whereas *GoCo-N* adds predictor-native neighbourhood evidence through the call-resolution path of the same section (Supplementary Note S3).

*Run time.* The reference implementation is vectorised over the certification fold. On a laptop CPU, all methods for 100 splits at three error targets and two values of  $\delta$  took 3 minutes for Wainberg, 13 minutes for Sanger and 5 minutes each for DRIVE and HAP1, dominated by the sparse source-attribution products.

*Out-of-sample informativeness of the ordering.* Across the certified blocks, GoCo's statistic  $u_i$  is rank-correlated with the realised increment  $\Delta_i$  on affected pool genes at Spearman 0.22–0.73, against  $-0.08$ – $0.43$  for the raw score of the added calls; the per-block values are in Table S4.

## S2.8 Source objects beyond co-essentiality

GoCo is defined for any predictor that exposes a pair  $(s_{ig}, \mathcal{R}_{ig})$ : a frozen score and, for each candidate call, the evidence objects that produced it. The distinction relevant to the candidate path is whether that evidence is shared across genes or assembled separately for each gene. Shared evidence can create exact tied blocks addressed by GoCo-M, whereas gene-specific evidence permits call-level ordering as in GoCo-N. Table S7 summarizes the source-object structures considered here, and Supplementary Note S3 develops the neighbourhood family on FunMap with validation on STRING. Every source object must be target-excluded: allowing the focal gene's own annotations to contribute to the evidence for the call later evaluated would inflate retrospective support. Corollary S1 ensures that the quality of the resulting ordering affects yield rather than validity.

## S2.9 Gene-level case study of the released annotations

In Wainberg split 0 GoCo-M admitted 40% of the affected genes and released 351 evaluation-fold gene–GO calls that GoDag did not, over 158 genes, of which 37 (10.5%) are supported by the frozen reference. Table S5 lists the eight co-essential modules that contributed the most of them; the 28 that contributed at least four are all in the shipped split-level output. The interleukin-6 signal-transduction module {IL6ST, JAK1, LIFR, SLC39A1, STAT3} transfers correctly: IL6ST and JAK1 each acquire interleukin-27- and interleukin-35-mediated signalling (GO:0070106, GO:0070757) from the other members of the receptor complex, and the reference supports all four calls. In the TNF-receptor-associated module {BIRC2, CYLD, N4BP1, TRAF2, TRAF3}, BIRC2 acquires the CD40 receptor complex (GO:0035631) from TRAF2 and TRAF3 and

regulation of the necroptotic process (GO:0060544) from CYLD, both supported. In the anaphase-promoting complex module {ANAPC5, ANAPC13, ANAPC15, UBE2C, UBE2S}, which contributes five admitted calls and so falls below the table's cut, three are supported, including GO:0005680 for ANAPC15 and UBE2C. Where a module mixes a regulator with its targets the transfer fails: CDKN1A, in {CDKN1A, TP53, TP53BP1, UBE2K, USP28}, acquires seven terms that belong to TP53 as a transcription factor—core promoter sequence-specific DNA binding, transcription cofactor binding, regulation of thymocyte apoptosis—and the reference supports none of them; DDX21 and NCL likewise each receive four nucleolar terms from NOLC1. Distinguishing these two regimes is exactly what the ordering is estimating, and it does so imperfectly but informatively (Table S4). The examples also show that a reference-unsupported call is not necessarily biologically false: N4BP1 acquires regulation of Toll-like receptor signalling (GO:0034121) from BIRC2, which the frozen reference does not carry, although N4BP1 is an established negative regulator of TLR-induced NF- $\kappa$ B signalling (Shi et al. 2021). Core promoter binding for CDKN1A, by contrast, illustrates an unsupported transfer of a regulator's function to its target; these retrospective data do not establish biological absence. Reference-supported yield is therefore a conservative measure of recovered biological support, while the measured gene-level FDP can overstate biological error relative to a more complete or context-specific annotation reference.

#### S2.10 References for this note

Shi H, Sun L, Wang Y, et al. N4BP1 negatively regulates NF- $\kappa$ B by binding and inhibiting NEMO oligomerization. *Nat Commun.* 2021;12:1379.

Wainberg M, Kamber RA, Balsubramani A, et al. A genome-wide atlas of co-essential modules assigns function to uncharacterized genes. *Nat Genet.* 2021;53:638–649.



### Supplementary Note S3: The neighbourhood-structure instantiation, its validation, and the choice of candidate path

This note reports the framework applied to a predictor of a different kind—a random-walk neighbourhood on a functional gene network rather than a co-essential module—together with validation without retuning on a second, unrelated network. It also records the converse: the candidate path that is best for this structure, GoCo-N, is *worse* than GoCo-M on Wainberg, and the decomposition of Lemma S6 says why the two structures call for different paths.

#### S3.1 FunMap: the predictor and the frozen objects

FunMap is a machine-learned pan-cancer functional gene network derived from CPTAC proteogenomics. We take the authors’ released network and their own top-50 random-walk-with-restart (RWR) neighbourhoods and retrain nothing. FunMap’s edges were learned from CPTAC proteomic and transcriptomic co-expression against a co-functionality gold standard drawn from Reactome, ProteomeHD, BioPlex, HuRI, HI-Union and BioGRID; Gene Ontology was not among its training labels or its edge scores and was used by its authors only to interpret modules after the fact. Because Reactome co-membership is correlated with GO Biological Process co-annotation, we treat FunMap as the development predictor of this family and use STRING for validation without retuning (Section S3.2). For a focal gene the wrapper removes that gene from its own neighbourhood and removes its annotations from the enrichment background, then runs hypergeometric over-representation over GO terms of size  $[10, 2000]$  with Benjamini–Hochberg adjustment within aspect; the score of a call is  $-\log_{10}$  of the adjusted  $p$ -value. The GO release is the 2024-09-08 archive, which predates the FunMap paper’s acceptance; the same frozen tables are used for STRING. The panel is the 10,316 GO-annotated genes of the 10,525-gene network, giving 3,303,702 scored gene–GO calls.

The predictor therefore exposes exactly the two objects the construction needs (Section 2.1 of the main text): a frozen score  $s_{ig}$ , and for each call a source set  $\mathcal{R}_{ig}$ , here the annotated neighbours carrying the term. One refinement is introduced by the structure. Wainberg’s module members are unordered, so Equation (S2) spreads each call’s unit of evidence uniformly over them; a random-walk neighbourhood is *ranked* by proximity, so the same attribution is weighted by  $w(\text{rank}) = 1/\log_2(1 + \text{rank})$ . Equation (S3) is used with this weighted attribution to construct the source-derived rate features; GoCo-N then orders individual calls by the fitted unsupported probabilities  $\hat{p}_{ig}$ .

*The per-call model.* A logistic regression of “call reference-unsupported” is fitted on ranking-fold calls and applied to every call. In the neighbourhood implementation, the two label-derived rate features, their shrinkage targets and the classifier coefficients are fitted on one hash-defined half of the ranking fold and used to score calls of the other half; every certification- or evaluation-fold call is scored by the model fitted on the whole ranking fold. This differs slightly from the module implementation described in Section S2.3, but both constructions use ranking-fold labels only. Splits of this family use seed 20260910 +  $s$  (Section S2.1). Its features are the neighbourhood analogues of the module features: from frozen predictor output, the log score, the number of annotated neighbours carrying the term, the number of annotated

neighbours, the call’s BH rank within its gene and aspect, the number of sources, their total proximity weight, the proximity rank of the closest source and the gene’s number of calls; from ranking-fold labels only, the proximity-weighted unsupported rate of the call’s sources and the term’s unsupported rate, each shrunk by two pseudo-calls. The enrichment’s background count for the term is deliberately *not* a feature. Because the focal gene is removed from the background, that count equals the term’s annotation count minus one exactly when the focal gene is itself Direct-annotated with the term, so it would encode the label being predicted; this is the same construction excluded from the Wainberg model for the same reason (Section S2.7). A draft of this analysis that included it gave GoCo-N yields about 2–4% higher than those reported here; every number in this paper is from the model without it.

#### S3.2 STRING: validation without retuning, and why this network and not another

Validation uses STRING v12.0 (human), restricted to the *experiments* evidence channel. Three properties support its use as an external validation here.

*Independent of the development predictor.* STRING v12.0 was frozen in July 2023, eighteen months before FunMap’s publication, and is built from physical-assay evidence (DIP, BioGRID, HPRD, IntAct, MINT, PDB), not from CPTAC proteogenomics. Neither network is an input to the other.

*Free of the annotations being evaluated.* This is the binding constraint, and it eliminates most candidate networks. GO enters STRING through two named columns, **database** (which lists Gene Ontology among its curated sources) and **textmining**; we read neither, using only column 10. The experiments channel is calibrated against KEGG and Complex Portal rather than GO. Selection cannot be GO-informed either: the minimum **combined\_score** over the 13,715,404 shipped rows is exactly 150 and **combined\_score**  $\geq$  **experiments**, so no pair with an experiments score above that floor can have been censored on GO-derived grounds. Networks whose edge weights are *fitted* to GO co-annotation—HumanNet, whose log-likelihood scores are trained against a GO Biological Process gold standard; HumanBase/GIANT, whose posteriors use co-annotation to propagated GO BP terms as the positive class; and GeneMANIA, whose integration weights are regressed on GO BP co-annotation—were excluded for this reason, since evaluating GO predictions against a network built from GO would be circular.

*Identical downstream pipeline.* At STRING’s published medium-confidence cut ( $\geq 400$ ) the experiments channel gives 92,885 undirected edges over 13,079 genes. Everything after the edge list is unchanged from FunMap: top-50 RWR neighbourhood, focal gene removed from the neighbourhood and from the background, hypergeometric over-representation on terms of size  $[10, 2000]$ , BH within aspect. Two facts must be declared rather than assumed. First, STRING publishes no neighbourhoods, so ours are reconstructed (restart 0.5, 50 power iterations); the validation therefore exercises the same predictor *recipe* rather than a second frozen author artefact, a weaker claim than FunMap’s. Second, we hold the GO truth fixed at the same frozen 2024-09-08 tables used for FunMap and restrict the analysis universe to the 8,496 genes those tables cover (the enrichment background is this universe minus the focal gene), of which 8,055 have a neighbourhood inside it and form the calibrated panel; the network is then the only thing that differs between the two applications. This costs

coverage relative to STRING’s full 13,079 genes and buys exact comparability of the reference. Our BH implementation reproduces FunMap’s published adjusted  $p$ -values on 97.6% of 3.3 million calls to within  $10^{-9}$ , which fixes the enrichment convention to the published one rather than to our reading of it.

### S3.3 Why the structure differs, and what Lemma S6 implies

Lemma S6 splits the value of partial admission inside a block into a *granularity* term—the ability to stop part-way through, worth something only because a global threshold must take the block whole—and an *ordering* term, the covariance  $(M/N) \text{Cov}_B\{\mathbf{1}(\text{rk}(i) \leq k), \Delta_i\}$ , which is negative when small-increment genes are admitted first. The two applications differ in which term is available.

In Wainberg the score is an enrichment statistic of three small integers, so 2,470 gene–GO pairs share the value 789.650 and the largest tie in the operating range holds 3,361. An exact tie has no interior threshold, so *no* global path, however fine, reaches the granularity term. Partial admission is necessary to reach it; GoCo-M uses gene-level admission within grid steps, whereas GoCo-N defines a separate path that orders individual calls by the model.

In FunMap and STRING the adjusted  $p$ -values carry no material ties in the operating range, so refining the global grid recovers much of the granularity available from threshold refinement at no cost in validity—the grid is a function of frozen scores alone. GoCo-N then uses the remaining predictor-native information to construct an individual-call ordering. This is a prespecified candidate-path heuristic motivated by the decomposition in Lemma S6, not an optimality result of that lemma. The candidate policies become the top- $\ell$  calls under the ranking-fold statistic—exactly  $\ell$  calls, ties in  $\hat{p}_{ig}$  being broken by the frozen gene hash and then by call index—for  $\ell$  on a fixed geometric grid of fractions of the call pool, nested in  $\ell$  and measurable with respect to  $\mathcal{T}$ , so Definition S1 is satisfied and main-text Proposition 1 and Corollary S1 apply without modification. This is the instantiation **GoCo-N** of Supplementary Note S4.1, fixed on FunMap; **GoCo-M** is the instantiation of Equation (S5), fixed on Wainberg. Both are admissible for every predictor; Section S3.4 compares them on FunMap and STRING, and Section S3.5 on the four shared-evidence data sets.

The two instantiations use different ranking resolutions. Equation (S4) ranks genes within a GoCo-M block by predicted loss increment per predicted supported term. GoCo-N instead defines a separate call-level heuristic that orders individual calls by  $\hat{p}_{ig}$ , with calls predicted most likely to be supported entering first. This probability ordering is not an algebraic specialisation of Equation (S4). The ranking model and certification framework are shared, but the candidate family and the resolution of the ordering differ.

### S3.4 What the comparison shows

The neighbourhood-structure panel of the main text gives the primary comparison, and Table S8 reports gains over the exhaustive matched-risk frontier of the global path—for each arm and split, the largest correct-call yield of any global threshold, over every distinct score value of the frozen score file and abstention, whose realised held-out risk does not exceed the risk the arm realised—isolating candidate-path quality from simply spending more of the available risk budget. Three points follow.

First, refining the global grid increased supported-call yield by approximately 5.0% at the primary  $\alpha = 0.10$ ,  $\delta = 0.50$  operating point in FunMap; at  $\delta = 0.10$  the corresponding increase was approximately 1.5%.

Second, GoCo-M—the instantiation fixed on Wainberg—transfers to this structure but captures only part of the available gain. Measured against the exhaustive matched-risk frontier at the risk each arm actually realised ( $\delta = 0.10$ ), GoCo-M gains +101, +196 and +164 supported calls at  $\alpha = 0.05, 0.10, 0.20$  in FunMap (ahead of the frontier in 100%, 100% and 100% of splits) and +46, +138 and +197 in STRING (100%, 100%, 100%).

Third, GoCo-N gains +155, +379 and +519 on the same frontier in FunMap (ahead in 95%, 100% and 100% of splits) and +172, +268 and +561 in STRING (100%, 97%, 99%), at realised risk comparable to GoCo-M’s (lower in 3 of the six combinations), and it widens coverage substantially: 1,084 against 779 annotated genes in FunMap at  $\alpha = 0.10$ . The pattern replicates on STRING at every target.

### S3.5 Both instantiations on the four dependency data sets

GoCo-N was also run on Wainberg, Sanger, DRIVE and HAP1 with the per-call model of Supplementary Note S4.1 unchanged—the same features and the same cross-fitting—so that only the candidate path differed. At  $\alpha = 0.10$ ,  $\delta = 0.10$ , GoCo-N released 142 fewer reference-supported terms than GoCo-M in Wainberg and 69 fewer in DRIVE, but 219 more in Sanger and 88 more in HAP1. Thus the prespecified structural mapping is a useful default rather than an oracle choice: even among predictors with shared evidence and exact ties, the better candidate path can vary. This motivates the ranking-fold selection rule of Section S3.6, which chooses between the two paths before certification without using evaluation-fold outcomes.

### S3.6 An adaptive alternative to the structural mapping

The main text fixes the instantiation by evidence structure, on one development predictor per structure. Because both paths satisfy Definition S1, any rule that chooses between them from  $\mathcal{T}$  and frozen inputs alone inherits main-text Proposition 1 unchanged (Corollary S1), so the choice can also be made per split at no cost in validity. The rule we report walks each instantiation on  $\mathcal{T}$  alone—the GoCo-M path with its ranking-fold cutoffs, the GoCo-N path over ranking-fold calls ordered by  $\hat{p}_{ig}$ —with the per-call model cross-fitted over ten hash-defined folds of  $\mathcal{T}$  using the family-specific feature construction described above, stops each at the last candidate whose ranking-fold mean loss is at most  $\alpha$ , and carries forward the instantiation with the larger ranking-fold supported-call yield.

Table S9 reports its verdict for every predictor and target beside the instantiation that in fact won on the evaluation fold. At  $\alpha = 0.10$  it chose GoCo-N in 0%, 96%, 18% and 89% of splits on Wainberg, Sanger, DRIVE and HAP1 and in 92% and 76% on FunMap and STRING—agreeing with the structural mapping on the two development predictors and departing from it on Sanger and HAP1—and it agreed with the held-out winner in 100%, 96%, 80%, 89%, 83% and 69% of splits. Where the two instantiations are far apart the rule is essentially never wrong; where they are closer it errs in a fraction of splits (31% on STRING at  $\alpha = 0.10$ , where GoCo-N’s held-out margin over GoCo-M is 13%), and the price is paid in power alone: what it selected released 6.0–73.4% more supported terms than the cutoff across the six predictors at

$\alpha = 0.10$ ,  $\delta = 0.10$ , against 6.3–23.3% for GoCo-M alone and -34.4–75.2% for GoCo-N alone.

We report the structural mapping in the main text because it is prespecified, needs no per-data-set decision and is what a user of a new predictor can apply immediately; the adaptive rule is available to a user who would rather spend ranking-fold labels on the choice. Predicting from a predictor's structure alone which instantiation wins, or by how much, remains open.

### S3.7 Measurability

The operational measurability test (`test_measurability.py`) was repeated for both applications: scrambling the TruePath labels of every gene outside the ranking fold leaves the per-call ordering bit-identical, over 241,154 pool calls in FunMap and 470,735 in STRING, with maximum absolute difference 0 in both, and likewise leaves the GoCo-M statistic  $u_i$  of every affected pool gene in every block unchanged. Source sets are drawn from neighbours' annotations, never the focal gene's: the FunMap neighbourhood file contains no self-loops in 499,261 rows, and the STRING neighbourhoods are constructed with the focal gene removed. The scrambling test detects any use of certification- or evaluation-fold labels through the truth tables; it cannot detect a label that a frozen predictor column carries by construction, which is why the background count is excluded by inspection rather than by test.

## Supplementary Note S4: Candidate construction, formal guarantees and data reconstruction

This note specifies the candidate construction and formal guarantees underlying the operational steps in the main Methods. Numbered citations refer to the main reference list.

### S4.1 Candidate construction

*Blocks and increments.* On the grid  $\lambda_1 > \dots > \lambda_J$  shared by GoDag and GoCo (Table S11), block  $j$  steps from conservative sets  $S_i^H = S_i(\lambda_j)$  to liberal sets  $S_i^L = S_i(\lambda_{j+1}) \supseteq S_i^H$ , and for gene  $i$

$$\Delta_i = L_i(S_i^L) - L_i(S_i^H), \quad c_i = |S_i^L| - |S_i^H|, \quad D_i = S_i^L \setminus S_i^H, \quad (S1)$$

with  $B_j = \{i : c_i > 0\}$  the *affected* genes,  $\mathcal{T}_j = B_j \cap \mathcal{T}$  the affected ranking genes and  $\phi_i = 1 - |D_i \cap A_i^{\text{TP}}|/c_i$  the unsupported fraction of the added calls.

*Fractional source attribution.* For each added call  $g \in D_i$  the predictor names the sources  $\mathcal{R}_{ig}$  behind it—in Wainberg the winning module's annotated members, gene  $i$  excluded—and each call spreads a unit of evidence equally over them:

$$a_{ir} = \frac{1}{c_i} \sum_{g \in D_i} \frac{\mathbf{1}\{r \in \mathcal{R}_{ig}\}}{|\mathcal{R}_{ig}|}, \quad i \in B_j. \quad (S2)$$

Every added call has a retained source in all four data sets (Supplementary Table S4).

*Source-level shrinkage.* For  $y_i = \Delta_i$  or  $y_i = \phi_i$ ,  $i \in \mathcal{T}_j$ , let  $\bar{y}$  denote the mean over the affected ranking genes. The source estimate is the attribution-weighted mean over the ranking genes  $r$  supports, shrunk toward  $\bar{y}$  by one pseudo-gene,

$$\hat{y}_r = \frac{\sum_{i \in \mathcal{T}_j} a_{ir} y_i + \kappa \bar{y}}{\sum_{i \in \mathcal{T}_j} a_{ir} + \kappa}, \quad \kappa = 1, \quad (S3)$$

and the prediction  $\hat{y}_i = \sum_r a_{ir} \hat{y}_r$  for an affected gene depends, outside  $\mathcal{T}$ , on losses only through ranking-fold genes: its own labels never enter.

*The GoCo statistic.* Admitting gene  $i$  changes pool risk by  $\Delta_i/N$  and gains  $\tau_i = |D_i \cap A_i^{\text{TP}}|$  supported terms. We therefore use  $\Delta_i/\tau_i$  as a cost-per-gain ordering heuristic, prioritising genes expected to add less gene-level FDP per supported term; this ordering is not claimed to solve the discrete allocation problem exactly, and optimality is not required for validity. Because both quantities are unknown for pool genes, GoCo estimates per-call probabilities  $\hat{p}_{ig}$  of being unsupported from ranking-fold information and uses them to construct  $\hat{\Delta}_i$  and  $\hat{\tau}_i$ . Its leading feature is the shrunk rate at which a call's sources  $\mathcal{R}_{ig}$  produced unsupported ranking-fold terms; all seven features come from ranking-fold labels or frozen output, so the fitted probabilities are  $\mathcal{T}$ -measurable (Supplementary Note S2.3). With  $\hat{L}_i(S) = |S|^{-1} \sum_{g \in S} \hat{p}_{ig}$  for  $|S| > 0$  and  $\hat{L}_i(\emptyset) = 0$ ,  $\hat{\Delta}_i = \hat{L}_i(S_i^L) - \hat{L}_i(S_i^H)$  and  $\hat{\tau}_i = \sum_{g \in D_i} (1 - \hat{p}_{ig})$ , GoCo orders affected genes by

$$u_i = \frac{\hat{\Delta}_i}{\max(\hat{\tau}_i, 10^{-3})}, \quad (S4)$$

the predicted loss increase per predicted supported term, so genes backed by reliable members and gaining several such terms go first. Any other  $\mathcal{T}$ -measurable ordering could be substituted without touching the certificate (Corollary S1).

*Partial admission.* Order  $B_j$  lexicographically by  $(u_i, h_i, \text{id}(i))$ , where  $h_i \in [0, 1)$  is a SHA-256 hash of the gene identifier and  $\text{id}(i)$  is the gene identifier used as a final deterministic tie-break. For  $q \in \mathcal{Q}$  with  $q < 1$  (Table S11), let  $(u^*, h^*, \text{id}^*)$  be the  $k(q)$ -th smallest triple among the affected ranking genes,  $k(q) = \max\{1, \lfloor q|\mathcal{T}_j| \rfloor\}$ , and set

$$Q_{j,q} = \{i \in B_j : (u_i, h_i, \text{id}(i)) \leq (u^*, h^*, \text{id}^*)\}, \quad (S5)$$

$$S_i(\pi_{j,q}) = \begin{cases} S_i^L, & i \in Q_{j,q}, \\ S_i^H, & i \notin Q_{j,q}. \end{cases}$$

For the endpoints we define  $Q_{j,0} = \emptyset$ , so  $\pi_{j,0}$  is the global-threshold policy at  $\lambda_j$  (already on the path, hence never inserted), and  $Q_{j,1} = B_j$ , so  $\pi_{j,1}$  is exactly the next global-threshold policy at  $\lambda_{j+1}$ . The cutoff is learned on  $\mathcal{T}$  and applied to every affected gene; candidate admission uses this deterministic total order, so the  $Q_{j,q}$  are nested, with  $Q_{j,1} = B_j$  corresponding to  $\lambda_{j+1}$  (Supplementary Note S1.4).

**Definition S1 (Candidate path)** A candidate path is a finite sequence of policies  $\pi_1, \dots, \pi_K$  that is (i) nested,  $S_i(\pi_k) \subseteq S_i(\pi_{k+1})$  for all  $i$  and  $k$ , and (ii)  $\mathcal{T}$ -measurable: the sequence, including its length  $K$ , is a function of the ranking fold (its members and their labels), the frozen inputs, and any auxiliary randomisation fixed before the split. All conditional statements below also condition on that fixed randomisation. Labels of  $\mathcal{C}$  and  $\mathcal{E}$  are never used.

GoDag's path is the label-free grid; GoCo-M's visits the same grid and, after  $\lambda_j$ , inserts  $\pi_{j,q}$  for  $q \in \mathcal{Q}$  in increasing order before  $\lambda_{j+1}$  whenever the ranking-fold mean loss at  $\lambda_{j+1}$  exceeds  $0.8\alpha$  and  $|\mathcal{T}_j| \geq 5$ . Both triggers use  $\mathcal{T}$  only, so every such path satisfies Definition S1 (Supplementary Lemma S4); the construction and its constants (Table S11) are identical across data sets.

GoCo-N instead orders individual calls by  $\hat{p}_{ig}$ ; its neighbourhood-specific features and the ranking-fold-only rule for choosing between paths are specified in Supplementary Note S3. Both constructions satisfy Definition S1.

#### S4.2 Certification rule and formal guarantees

For candidate  $\pi_k$  define

$$\hat{R}_k = \frac{1}{m} \sum_{i \in \mathcal{C}} L_i(\pi_k), \quad \hat{\sigma}_k^2 = \frac{1}{m} \sum_{i \in \mathcal{C}} \{L_i(\pi_k) - \hat{R}_k\}^2, \quad (\text{S6})$$

$$p_k = \Phi\left(\frac{\hat{R}_k - \alpha}{\max(\hat{\sigma}_k, 10^{-6})/\sqrt{m}}\right), \quad (\text{S7})$$

the one-sided normal  $p$ -value of the released Multilabel implementation [24] (default  $\delta = 0.50$ ), used unchanged by GoDag and GoCo on the hierarchy-aware loss. In the module-family analyses, GoDag and the other global methods use  $\mathcal{T} \cup \mathcal{C}$  and  $n_{\mathcal{T}} + m$  in place of  $\mathcal{C}$  and  $m$ ; in the neighbourhood-family analyses, all compared methods certify on  $\mathcal{C}$ . Fixed-sequence Learn-Then-Test [23] tests  $p_1, p_2, \dots$  in path order against  $H_k : R(\pi_k) \geq \alpha$ , stops at the first  $k$  with  $p_k > \delta$  and returns the preceding candidate  $\hat{\pi}$  (abstention, risk zero, never occurred). Here  $\delta$  is the tolerated false-certification level in the asymptotic guarantee in main-text Proposition 1; smaller  $\delta$  requires stronger evidence before a candidate is certified. Given  $\mathcal{T}$  the path is fixed and  $\mathcal{C}$  a random sample of the pool, so every  $\hat{R}_k$  is an unbiased survey estimate.

The conditional fixed-sequence guarantee is stated in main-text Proposition 1, with its proof in Supplementary Note S1.6. The argument also covers global normal-test methods on their respective certification folds. Replacing the normal test with Hoeffding’s bound [34] gives finite-sample control on the same prespecified path.

**Corollary S1** (Validity does not depend on the ordering) *Under the assumptions of main-text Proposition 1, validity does not depend on how the  $\mathcal{T}$ -measurable ordering is constructed. It therefore holds for any ordering of the affected genes, or of individual calls, that is a function of the ranking fold and frozen inputs—orderings based only on frozen inputs, such as a deterministic hash or the score of the added calls, Equation (S4), and the GoCo-N path—with or without the insertion triggers, and for a choice between such paths made on  $\mathcal{T}$ . Predictor-native evidence changes which policies are reached before the first failure, not the error bound.*

**Remark S2** (The  $\delta$  identity and held-out exceedance) *Certification is equivalent to  $\hat{R}_k \leq \alpha + z_{\delta} \max(\hat{\sigma}_k, 10^{-6})/\sqrt{m}$ . Thus, at  $\delta = 0.50$ , normal-type procedures with a positive scale estimate reduce to  $\hat{R}_k \leq \alpha$ . Multilabel, IID-normal and network-HAC therefore select the same policy on a common path and loss; this cancellation does not apply to concentration-bound tests. Pool failure and held-out exceedance remain different events. Supplementary Notes S1.7 and S2.4 give the exact split identity, the rounding factors and the stricter  $\delta = 0.10$  calculation.*

Validity does not depend on the ordering; yield does, through two separable effects.

**Lemma S6** (Granularity and the value of an ordering) *Fix  $\mathcal{T}$  and a block, let  $B = B_j \cap \mathcal{P}$  with  $M = |B| \geq 1$  and block*

*mean  $\bar{\Delta}_B = M^{-1} \sum_{i \in B} \Delta_i$ . For a  $\mathcal{T}$ -measurable strict order on  $B$ , let  $\text{rk}(i) \in \{1, \dots, M\}$  be the position of gene  $i$ , let  $\pi_{(k)}$  relax exactly the  $k$  first-ranked genes, and let  $\pi_{(k)}^{\text{rand}}$  relax a uniformly random  $k$ -subset of  $B$ .*

- (a) Granularity.  $R(\pi_{(k)}) = R(S^H) + N^{-1} \sum_{\text{rk}(i) \leq k} \Delta_i$  and  $\mathbb{E} R(\pi_{(k)}^{\text{rand}}) = R(S^H) + (k/M) \{R(S^L) - R(S^H)\}$ . Random partial admission interpolates linearly between the two global endpoints, whereas a global threshold realises at most  $r - 1$  intermediate policies when the block contains  $r$  distinct score values, and none for an exact tie.
- (b) Ordering value.  $R(\pi_{(k)}) - \mathbb{E} R(\pi_{(k)}^{\text{rand}}) = (M/N) \text{Cov}_B \{\mathbf{1}(\text{rk}(i) \leq k), \Delta_i\}$ , the empirical covariance over  $B$ . A useful ordering admits small- $\Delta$  genes first, making this covariance negative. Averaged over  $k = 1, \dots, M$ , the risk saving of the ordering relative to random admission equals  $N^{-1} \text{Cov}_B \{\text{rk}(i), \Delta_i\}$ ; it is zero in expectation for a uniformly random ordering independent of the increments, and if  $\Delta_i$  is nondecreasing in  $\text{rk}(i)$  the ordering is optimal among all orderings at every  $k$ .

*Proof idea.* (a) is linearity of the pool mean; (b) follows from  $\sum_k \mathbf{1}\{\text{rk}(i) \leq k\} = M + 1 - \text{rk}(i)$  and Chebyshev’s sum inequality (Supplementary Note S1.8).

#### S4.3 Module reconstruction and frozen reference

Wainberg et al. built overlapping human co-essential modules from genome-scale CRISPR dependency profiles and assigned functions by leave-one-gene-out GO enrichment of the module [19]. Our reconstruction reproduces the archived Wainberg score file used in this analysis to numerical precision and names, for every row, the winning non-syntenic module and the annotated members carrying the signal: those members are the sources  $\mathcal{R}_{ig}$  (panel 15,794 genes). The Direct reference and its TruePath closure were fixed before analysis (284,700 and 1,851,997 pairs over 19,578 genes; GO release of 1 November 2019, all evidence codes, NOT-qualified annotations removed, closure over *is\_a* and *part\_of*), and every method uses the same 100 split streams, score matrices and held-out truth. Sanger Project Score [35] and Project DRIVE [36] add CRISPR and RNAi evidence, choosing among the same Wainberg modules by correlation with the leave-target-out centroid—no GO information enters that choice—before the same target-excluded enrichment rule (14,885 and 6,283 eligible genes); HAP1 uses the published Global-L2 parent clusters of its genetic-interaction network [37] (15,020 genes). All three share the gene universe and GO reference with Wainberg; only the dependency evidence changes. The neighbourhood-structure family—FunMap random-walk neighbourhoods [38] and the experiments channel of STRING v12.0 [39]—is described in Supplementary Note S3.

#### Supplementary Note S5: Declaration of artificial intelligence assistance

*Declaration of the use of artificial intelligence.* The authors used Claude Opus 5 (Anthropic) and ChatGPT GPT-5.6 Sol (OpenAI) as aids for language editing of author-written text, consistency checking, and assistance with code writing and documentation. All scientific decisions, interpretations and conclusions were made by the authors. All model-assisted suggestions and code were reviewed and verified by the authors, who take full responsibility for the manuscript.

**Table S1.** Operating characteristics of Multilabel, GoDag, GoCo-M, GoCo-N and  $\mathcal{T}$ -selected GoCo at  $\alpha = 0.10$  and  $\delta \in \{0.50, 0.10\}$ , over 100 splits.

| $\delta$ | Data set | Method                                | Gene FDP | $\Pr(\widehat{R}_{\mathcal{E}} > \alpha)$ | $\Pr(R > \alpha)$ | Genes w/ correct call | Calls | Correct calls | $\Delta$ correct calls vs GoDag | $\Pr(> \text{GoDag})$ |
|----------|----------|---------------------------------------|----------|-------------------------------------------|-------------------|-----------------------|-------|---------------|---------------------------------|-----------------------|
| 0.50     | Wainberg | Multilabel (Direct loss)              | 0.0923   | 0.01                                      | 0.00              | 185.7                 | 1614  | 339.6         | +0.0                            | –                     |
|          |          | GoDag                                 | 0.0923   | 0.01                                      | 0.00              | 185.7                 | 1614  | 339.6         | –                               | –                     |
|          |          | <b>GoCo-M</b>                         | 0.0964   | 0.20                                      | 0.08              | 197.1                 | 1955  | 369.1         | +29.4                           | 1.00                  |
|          |          | <b>GoCo-N</b>                         | 0.0944   | 0.16                                      | 0.07              | 129.3                 | 1208  | 227.1         | –112.5                          | 0.01                  |
|          |          | <b>GoCo</b> , $\mathcal{T}$ -selected | 0.0964   | 0.20                                      | 0.08              | 197.1                 | 1955  | 369.1         | +29.4                           | 1.00                  |
|          | Sanger   | Multilabel (Direct loss)              | 0.0953   | 0.20                                      | 0.00              | 155.4                 | 1919  | 333.0         | –20.8                           | –                     |
|          |          | GoDag                                 | 0.0980   | 0.36                                      | 0.00              | 160.9                 | 2040  | 353.8         | –                               | –                     |
|          |          | <b>GoCo-M</b>                         | 0.0980   | 0.42                                      | 0.20              | 166.9                 | 2380  | 380.7         | +26.9                           | 0.81                  |
|          |          | <b>GoCo-N</b>                         | 0.0942   | 0.11                                      | 0.03              | 206.9                 | 1905  | 572.2         | +218.4                          | 1.00                  |
|          |          | <b>GoCo</b> , $\mathcal{T}$ -selected | 0.0943   | 0.11                                      | 0.04              | 205.7                 | 1916  | 567.2         | +213.4                          | 0.99                  |
|          | DRIVE    | Multilabel (Direct loss)              | 0.0945   | 0.13                                      | 0.06              | 96.1                  | 1834  | 248.1         | –10.5                           | –                     |
|          |          | GoDag                                 | 0.0965   | 0.28                                      | 0.28              | 98.7                  | 1954  | 258.6         | –                               | –                     |
|          |          | <b>GoCo-M</b>                         | 0.0971   | 0.33                                      | 0.26              | 105.7                 | 2296  | 291.7         | +33.1                           | 0.89                  |
|          |          | <b>GoCo-N</b>                         | 0.0984   | 0.36                                      | 0.25              | 78.3                  | 1417  | 212.0         | –46.6                           | 0.12                  |
|          |          | <b>GoCo</b> , $\mathcal{T}$ -selected | 0.0974   | 0.34                                      | 0.25              | 100.6                 | 2160  | 277.5         | +18.9                           | 0.72                  |
|          | HAP1     | Multilabel (Direct loss)              | 0.0719   | 0.00                                      | 0.00              | 77.0                  | 847   | 129.4         | +0.0                            | –                     |
|          |          | GoDag                                 | 0.0719   | 0.00                                      | 0.00              | 77.0                  | 847   | 129.4         | –                               | –                     |
|          |          | <b>GoCo-M</b>                         | 0.0943   | 0.19                                      | 0.05              | 80.8                  | 1285  | 137.7         | +8.2                            | 1.00                  |
|          |          | <b>GoCo-N</b>                         | 0.0931   | 0.17                                      | 0.05              | 110.5                 | 997   | 228.7         | +99.3                           | 1.00                  |
|          |          | <b>GoCo</b> , $\mathcal{T}$ -selected | 0.0929   | 0.15                                      | 0.05              | 107.7                 | 1029  | 220.9         | +91.5                           | 1.00                  |
| 0.10     | Wainberg | Multilabel (Direct loss)              | 0.0923   | 0.01                                      | 0.00              | 185.3                 | 1612  | 338.9         | –0.7                            | –                     |
|          |          | GoDag                                 | 0.0923   | 0.01                                      | 0.00              | 185.7                 | 1614  | 339.6         | –                               | –                     |
|          |          | <b>GoCo-M</b>                         | 0.0948   | 0.13                                      | 0.00              | 195.2                 | 1896  | 365.2         | +25.6                           | 0.98                  |
|          |          | <b>GoCo-N</b>                         | 0.0899   | 0.05                                      | 0.00              | 127.0                 | 1150  | 222.8         | –116.8                          | 0.00                  |
|          |          | <b>GoCo</b> , $\mathcal{T}$ -selected | 0.0948   | 0.13                                      | 0.00              | 195.2                 | 1896  | 365.2         | +25.6                           | 0.98                  |
|          | Sanger   | Multilabel (Direct loss)              | 0.0913   | 0.01                                      | 0.00              | 150.7                 | 1751  | 315.8         | –3.3                            | –                     |
|          |          | GoDag                                 | 0.0923   | 0.12                                      | 0.00              | 151.7                 | 1789  | 319.1         | –                               | –                     |
|          |          | <b>GoCo-M</b>                         | 0.0930   | 0.15                                      | 0.00              | 156.9                 | 1983  | 340.0         | +20.9                           | 0.95                  |
|          |          | <b>GoCo-N</b>                         | 0.0910   | 0.04                                      | 0.00              | 202.7                 | 1820  | 559.2         | +240.1                          | 1.00                  |
|          |          | <b>GoCo</b> , $\mathcal{T}$ -selected | 0.0910   | 0.04                                      | 0.00              | 201.3                 | 1824  | 553.4         | +234.3                          | 1.00                  |
|          | DRIVE    | Multilabel (Direct loss)              | 0.0913   | 0.13                                      | 0.00              | 90.0                  | 1609  | 227.9         | –9.3                            | –                     |
|          |          | GoDag                                 | 0.0929   | 0.13                                      | 0.00              | 92.9                  | 1718  | 237.2         | –                               | –                     |
|          |          | <b>GoCo-M</b>                         | 0.0931   | 0.16                                      | 0.01              | 99.4                  | 1987  | 263.5         | +26.2                           | 0.89                  |
|          |          | <b>GoCo-N</b>                         | 0.0923   | 0.18                                      | 0.01              | 73.3                  | 1145  | 194.5         | –42.8                           | 0.17                  |
|          |          | <b>GoCo</b> , $\mathcal{T}$ -selected | 0.0931   | 0.17                                      | 0.01              | 94.5                  | 1850  | 251.5         | +14.3                           | 0.75                  |
|          | HAP1     | Multilabel (Direct loss)              | 0.0719   | 0.00                                      | 0.00              | 77.0                  | 847   | 129.4         | +0.0                            | –                     |
|          |          | GoDag                                 | 0.0719   | 0.00                                      | 0.00              | 77.0                  | 847   | 129.4         | –                               | –                     |
|          |          | <b>GoCo-M</b>                         | 0.0915   | 0.06                                      | 0.00              | 80.7                  | 1252  | 137.6         | +8.1                            | 1.00                  |
|          |          | <b>GoCo-N</b>                         | 0.0895   | 0.05                                      | 0.00              | 108.4                 | 965   | 225.1         | +95.7                           | 1.00                  |
|          |          | <b>GoCo</b> , $\mathcal{T}$ -selected | 0.0893   | 0.04                                      | 0.00              | 105.7                 | 996   | 217.5         | +88.0                           | 1.00                  |

Means over the same 100 split streams at  $\alpha = 0.10$ . Gene FDP: held-out gene-level TruePath FDP  $\widehat{R}_{\mathcal{E}}$ .  $\Pr(\widehat{R}_{\mathcal{E}} > \alpha)$ : fraction of splits whose held-out FDP exceeded  $\alpha$ .  $\Pr(R > \alpha)$ : fraction of splits in which the TruePath risk of the certified policy over the pool of its own certificate (the whole panel for Multilabel and GoDag,  $\mathcal{C} \cup \mathcal{E}$  for GoCo) exceeded  $\alpha$ ; for GoDag and GoCo this is the event main-text Proposition 1 bounds. Multilabel certifies the Direct loss instead: on that loss its pool failure frequencies were 0.00/0.13/0.06/0.00 (Wainberg/Sanger/DRIVE/HAP1) at  $\delta = 0.50$  and 0.00/0.00/0.00/0.00 at  $\delta = 0.10$ , and a policy certified on the stricter Direct loss is conservative on TruePath. Correct calls: released gene–GO pairs the frozen reference confirms; genes w/ correct call: held-out genes receiving at least one of them; Calls: released calls of any kind. Multilabel is the released multilabel calibration of Boger et al. on the Direct loss, calibrated on 70% of the panel; GoDag and GoCo-M use the TruePath loss and share the 25-unit grid; GoCo-N walks the top- $\ell$  calls of the ranking-fold model; GoCo,  $\mathcal{T}$ -selected is whichever of the two the ranking-fold rule chose in that split (Table S9).  $\Delta$  correct calls: paired mean difference in correct calls from GoDag;  $\Pr(> \text{GoDag})$ : fraction of splits in which the method released more correct calls than GoDag.



**Table S2.** Error-target sweep at  $\delta = 0.10$ : mean held-out gene-level TruePath FDP and reference-supported calls at  $\alpha \in \{0.05, 0.10, 0.20\}$  over 100 splits.

| Data set                                                     | Method                                | $\alpha = 0.05$ |               | $\alpha = 0.10$ |               | $\alpha = 0.20$ |               |
|--------------------------------------------------------------|---------------------------------------|-----------------|---------------|-----------------|---------------|-----------------|---------------|
|                                                              |                                       | FDP             | Correct calls | FDP             | Correct calls | FDP             | Correct calls |
| <i>Module structure: GoCo-M (development: Wainberg)</i>      |                                       |                 |               |                 |               |                 |               |
| Wainberg                                                     | Multilabel (Direct loss)              | 0.0359          | 183.1         | 0.0923          | 338.9         | 0.1699          | 518.9         |
|                                                              | GoDag                                 | 0.0359          | 183.1         | 0.0923          | 339.6         | 0.1699          | 518.9         |
|                                                              | <b>GoCo-M</b>                         | 0.0438          | 213.2         | 0.0948          | 365.2         | 0.1882          | 539.2         |
| Sanger                                                       | Multilabel (Direct loss)              | 0.0450          | 198.2         | 0.0913          | 315.8         | 0.1871          | 1136.0        |
|                                                              | GoDag                                 | 0.0461          | 203.9         | 0.0923          | 319.1         | 0.1895          | 1163.2        |
|                                                              | <b>GoCo-M</b>                         | 0.0463          | 213.7         | 0.0930          | 340.0         | 0.1907          | 1346.8        |
| DRIVE                                                        | Multilabel (Direct loss)              | 0.0413          | 90.0          | 0.0913          | 227.9         | 0.1823          | 691.4         |
|                                                              | GoDag                                 | 0.0414          | 90.1          | 0.0929          | 237.2         | 0.1844          | 702.4         |
|                                                              | <b>GoCo-M</b>                         | 0.0436          | 94.2          | 0.0931          | 263.5         | 0.1841          | 702.0         |
| HAP1                                                         | Multilabel (Direct loss)              | 0.0436          | 96.0          | 0.0719          | 129.4         | 0.1353          | 160.2         |
|                                                              | GoDag                                 | 0.0436          | 96.1          | 0.0719          | 129.4         | 0.1353          | 160.2         |
|                                                              | <b>GoCo-M</b>                         | 0.0450          | 102.4         | 0.0915          | 137.6         | 0.1756          | 187.7         |
| <i>Neighbourhood structure: GoCo-N (development: FunMap)</i> |                                       |                 |               |                 |               |                 |               |
| FunMap                                                       | Multilabel (Direct loss, shared grid) | 0.0312          | 727.9         | 0.0583          | 1004.3        | 0.1543          | 1869.5        |
|                                                              | GoDag, shared grid                    | 0.0312          | 727.9         | 0.0901          | 1261.2        | 0.1547          | 1873.0        |
|                                                              | GoDag, refined grid                   | 0.0448          | 869.7         | 0.0941          | 1279.8        | 0.1900          | 2164.4        |
|                                                              | <b>GoCo-N</b>                         | 0.0431          | 1016.1        | 0.0878          | 1615.8        | 0.1823          | 2615.8        |
| STRING                                                       | Multilabel (Direct loss, shared grid) | 0.0385          | 423.2         | 0.0822          | 815.8         | 0.1458          | 1383.8        |
|                                                              | GoDag, shared grid                    | 0.0386          | 424.9         | 0.0828          | 828.1         | 0.1464          | 1387.9        |
|                                                              | GoDag, refined grid                   | 0.0454          | 473.1         | 0.0928          | 909.2         | 0.1867          | 1711.7        |
|                                                              | <b>GoCo-N</b>                         | 0.0429          | 621.4         | 0.0894          | 1151.1        | 0.1830          | 2243.0        |

FDP: mean held-out gene-level TruePath false discovery proportion. Correct calls: released gene-GO pairs the frozen reference confirms. Each structure is shown with the instantiation prespecified for it. Module-family GoCo-M and GoDag share the 25-unit grid. In the neighbourhood family, Multilabel and GoDag are shown on the shared coarse grid, with refined-grid GoDag separately; GoCo-N uses a distinct call-ordered path. Grid refinement does not change the fixed-sequence multiplicity argument.

**Table S3.** Certification, pool and held-out risks with yield diagnostics for GoDag and GoCo-M at  $\alpha = 0.10$  and  $\delta \in \{0.50, 0.10\}$ , averaged over 100 splits. GoCo-M certifies on  $\mathcal{C}$  (60%); GoDag calibrates on  $\mathcal{T} \cup \mathcal{C}$  (70%).

| $\delta$ | Data set | Method | Calib. mean | $R$    | $\hat{R}_E$ | $\Pr(R > \alpha)$ | $\Pr(\hat{R}_E > \alpha)$ | Genes w/ calls | Calls | Correct calls | Terms per supp. gene |
|----------|----------|--------|-------------|--------|-------------|-------------------|---------------------------|----------------|-------|---------------|----------------------|
| 0.50     | Wainberg | GoDag  | 0.0924      | 0.0924 | 0.0923      | 0.00              | 0.01                      | 561            | 1614  | 339.6         | 1.83                 |
| 0.50     | Wainberg | GoCo-M | 0.0964      | 0.0964 | 0.0964      | 0.08              | 0.20                      | 575            | 1955  | 369.1         | 1.87                 |
| 0.50     | Sanger   | GoDag  | 0.0984      | 0.0983 | 0.0980      | 0.00              | 0.36                      | 530            | 2040  | 353.8         | 2.20                 |
| 0.50     | Sanger   | GoCo-M | 0.0986      | 0.0984 | 0.0980      | 0.20              | 0.42                      | 529            | 2380  | 380.7         | 2.28                 |
| 0.50     | DRIVE    | GoDag  | 0.0955      | 0.0958 | 0.0965      | 0.28              | 0.28                      | 215            | 1954  | 258.6         | 2.62                 |
| 0.50     | DRIVE    | GoCo-M | 0.0962      | 0.0965 | 0.0971      | 0.26              | 0.33                      | 215            | 2296  | 291.7         | 2.76                 |
| 0.50     | HAP1     | GoDag  | 0.0720      | 0.0720 | 0.0719      | 0.00              | 0.00                      | 377            | 847   | 129.4         | 1.68                 |
| 0.50     | HAP1     | GoCo-M | 0.0939      | 0.0941 | 0.0943      | 0.05              | 0.19                      | 475            | 1285  | 137.7         | 1.70                 |
| 0.10     | Wainberg | GoDag  | 0.0924      | 0.0924 | 0.0923      | 0.00              | 0.01                      | 561            | 1614  | 339.6         | 1.83                 |
| 0.10     | Wainberg | GoCo-M | 0.0946      | 0.0947 | 0.0948      | 0.00              | 0.13                      | 568            | 1896  | 365.2         | 1.87                 |
| 0.10     | Sanger   | GoDag  | 0.0925      | 0.0924 | 0.0923      | 0.00              | 0.12                      | 502            | 1789  | 319.1         | 2.10                 |
| 0.10     | Sanger   | GoCo-M | 0.0933      | 0.0932 | 0.0930      | 0.00              | 0.15                      | 505            | 1983  | 340.0         | 2.17                 |
| 0.10     | DRIVE    | GoDag  | 0.0918      | 0.0921 | 0.0929      | 0.00              | 0.13                      | 208            | 1718  | 237.2         | 2.55                 |
| 0.10     | DRIVE    | GoCo-M | 0.0921      | 0.0925 | 0.0931      | 0.01              | 0.16                      | 208            | 1987  | 263.5         | 2.64                 |
| 0.10     | HAP1     | GoDag  | 0.0720      | 0.0720 | 0.0719      | 0.00              | 0.00                      | 377            | 847   | 129.4         | 1.68                 |
| 0.10     | HAP1     | GoCo-M | 0.0912      | 0.0913 | 0.0915      | 0.00              | 0.06                      | 462            | 1252  | 137.6         | 1.70                 |

**Table S4.** Block diagnostics at  $\alpha = 0.10$  for grid steps in which GoCo-M policies were certified, averaged over 100 splits. Spearman correlations compare the realised loss increment  $\Delta_i$  with the GoCo ordering statistic  $u_i$  and with the negative mean added-call score.

| Data set | Step                  | Affected | Ranking | mean $\Delta$ | sd $\Delta$ | full mass | $\rho(u, \Delta)$ | $\rho(-\text{score}, \Delta)$ | $\Pr(\rho_u > 0)$ |
|----------|-----------------------|----------|---------|---------------|-------------|-----------|-------------------|-------------------------------|-------------------|
| Wainberg | 800 $\rightarrow$ 775 | 1267     | 127     | 0.591         | 0.484       | 1.00      | 0.725             | -0.077                        | 1.00              |
| Sanger   | 450 $\rightarrow$ 425 | 454      | 45      | 0.203         | 0.408       | 1.00      | 0.560             | 0.041                         | 1.00              |
| Sanger   | 425 $\rightarrow$ 400 | 211      | 21      | 0.097         | 0.316       | 1.00      | 0.325             | 0.146                         | 1.00              |
| Sanger   | 400 $\rightarrow$ 375 | 1164     | 117     | 0.163         | 0.363       | 1.00      | 0.496             | 0.020                         | 1.00              |
| DRIVE    | 275 $\rightarrow$ 250 | 419      | 41      | 0.066         | 0.237       | 1.00      | 0.222             | 0.120                         | 1.00              |
| DRIVE    | 250 $\rightarrow$ 225 | 503      | 50      | 0.103         | 0.305       | 1.00      | 0.318             | 0.048                         | 1.00              |
| HAP1     | 475 $\rightarrow$ 450 | 1021     | 102     | 0.762         | 0.421       | 1.00      | 0.703             | 0.433                         | 1.00              |

**Table S5.** Additional calls released by GoCo-M but not GoDag in Wainberg split 0 at  $\alpha = 0.10$ ,  $\delta = 0.50$  and  $q = 0.4$ . Shown are the eight co-essential modules contributing the most additional calls; \* marks terms supported by the frozen TruePath reference.

| Module | Members                                        | Genes admitted      | Calls | Correct | Terms transferred                                                                                                                                                                                                                                                                                                                                                                     |
|--------|------------------------------------------------|---------------------|-------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5071   | CHEK2, MDM2, PPM1D, TP53, USP7                 | MDM2, TP53, USP7    | 18    | 2       | GO:0001046 core promoter sequence-specific DNA binding; GO:0001221 transcription cofactor binding; GO:0002309 T cell proliferation involved in immune response; GO:0010216 maintenance of DNA methylation; GO:0070245 positive regulation of thymocyte apoptotic process; GO:0071494 cellular response to UV-C *; GO:1901525 negative regulation of mitophagy; ...                    |
| 792    | CDS2, CHMP3, CHMP4B, INPP5A, VRK1              | CDS2, CHMP3, INPP5A | 11    | 1       | GO:0000815 ESCRT III complex; GO:0090148 membrane fission; GO:0090611 ubiquitin-independent protein catabolic process via the multivesicular body sorting pathway; GO:1902188 positive regulation of viral release from host cell; GO:0050792 regulation of viral process *                                                                                                           |
| 4493   | CDC42, CTNNA1, MLLT4, PARD6B, RAPGEF2, TP53BP2 | CTNNA1, TP53BP2     | 11    | 0       | GO:0003161 cardiac conduction system development; GO:0017034 Rap guanyl-nucleotide exchange factor activity; GO:0021884 forebrain neuron development; GO:0060501 positive regulation of epithelial cell proliferation involved in lung morphogenesis; GO:2000481 positive regulation of cAMP-dependent protein kinase activity; GO:0071681 cellular response to indole-3-methanol     |
| 3178   | DDX21, DNAJC2, HSPA14, NCL, PINX1              | DDX21, DNAJC2, NCL  | 11    | 0       | GO:0006450 regulation of translational fidelity; GO:0044547 DNA topoisomerase binding; GO:1902570 protein localization to nucleolus; GO:1904751 positive regulation of protein localization to nucleolus                                                                                                                                                                              |
| 4091   | BIRC2, CYLD, N4BP1, TRAF2, TRAF3               | BIRC2, N4BP1, TRAF2 | 9     | 4       | GO:0031996 thioesterase binding; GO:0035631 CD40 receptor complex *; GO:0060544 regulation of necroptotic process *; GO:0097300 programmed necrotic cell death *; GO:0034121 regulation of toll-like receptor signaling pathway; GO:1902527 positive regulation of protein monoubiquitination                                                                                         |
| 3710   | IL6ST, JAK1, LIFR, SLC39A1, STAT3              | IL6ST, JAK1         | 8     | 4       | GO:0034112 positive regulation of homotypic cell-cell adhesion; GO:0060019 radial glial cell differentiation; GO:0070106 interleukin-27-mediated signaling pathway *; GO:0070757 interleukin-35-mediated signaling pathway *; GO:0150105 protein localization to cell-cell junction                                                                                                   |
| 5036   | CDKN1A, TP53, TP53BP1, UBE2K, USP28            | CDKN1A, TP53        | 8     | 0       | GO:0001046 core promoter sequence-specific DNA binding; GO:0001221 transcription cofactor binding; GO:0002309 T cell proliferation involved in immune response; GO:0010994 free ubiquitin chain polymerization; GO:0070245 positive regulation of thymocyte apoptotic process; GO:1901525 negative regulation of mitophagy; GO:1902749 regulation of cell cycle G2/M phase transition |
| 4150   | DDX21, HSPA14, NCL, NOLC1, PINX1               | DDX21, NCL          | 8     | 0       | GO:0001042 RNA polymerase I core binding; GO:0007000 nucleolus organization; GO:0031428 box C/D snoRNP complex; GO:0042306 regulation of protein import into nucleus                                                                                                                                                                                                                  |

**Table S6.** Comparison of pooled call-level FDR and gene-level FDP estimands at  $\alpha = 0.10$ ,  $\delta = 0.50$ , over the same 100 splits.

| Data set | Procedure / target                    | Calls released | Correct calls | Realised pooled call FDP | Genes annotated |
|----------|---------------------------------------|----------------|---------------|--------------------------|-----------------|
| Wainberg | Conformal selection (pooled-call FDR) | 0.0            | 0.0           | 0.000                    | 0.0             |
|          | GoCo-M (gene-level FDP target)        | 1955           | 369.1         | 0.811                    | 575             |
| Sanger   | Conformal selection (pooled-call FDR) | 5.6            | 4.8           | 0.045                    | 5.4             |
|          | GoCo-M (gene-level FDP target)        | 2380           | 380.7         | 0.839                    | 529             |
| DRIVE    | Conformal selection (pooled-call FDR) | 3.6            | 2.8           | 0.078                    | 2.5             |
|          | GoCo-M (gene-level FDP target)        | 2296           | 291.7         | 0.872                    | 215             |
| HAP1     | Conformal selection (pooled-call FDR) | 0.2            | 0.1           | 0.006                    | 0.1             |
|          | GoCo-M (gene-level FDP target)        | 1285           | 137.7         | 0.893                    | 475             |

**Table S7.** Source-object structures, observed score granularity and ranking-fold path choice. Rows without a ranking-fold verdict are structural examples.

| Predictor family                                                         | Source set $\mathcal{R}_{ig}$               | Evidence      | Score ties                | Structure     | A cutoff misses                             | Ranking-fold verdict                                               |
|--------------------------------------------------------------------------|---------------------------------------------|---------------|---------------------------|---------------|---------------------------------------------|--------------------------------------------------------------------|
| Co-essential modules (Wainberg; Sanger and DRIVE via the same catalogue) | annotated members of the gene's module      | shared        | exact, up to 3,361 pairs  | module-shared | the interior of every tied block            | Wainberg: GoCo-M (100%); Sanger: GoCo-N (96%); DRIVE: GoCo-M (82%) |
| Protein complexes (CORUM, BioPlex)                                       | annotated members of the complex            | shared        | possible exact            | module-shared | the interior of exact ties                  | structural                                                         |
| Co-regulation clusters (ProteomeHD)                                      | annotated members of the co-regulated group | shared        | possible exact            | module-shared | the interior of exact ties                  | structural                                                         |
| Community / partition detection (HAP1 Global-L2 clusters)                | annotated members of the community          | shared        | exact, up to 14,245 pairs | module-shared | the interior of every tied block            | HAP1: GoCo-N (89%)                                                 |
| Random-walk neighbourhoods (FunMap)                                      | annotated neighbours, proximity-ranked      | gene-specific | none material             | neighbourhood | little: a refined grid recovers granularity | FunMap: GoCo-N (92%)                                               |
| Interaction neighbourhoods (STRING)                                      | annotated interactors, ranked               | gene-specific | none material             | neighbourhood | little: a refined grid recovers granularity | STRING: GoCo-N (76%)                                               |
| Network propagation / diffusion                                          | diffusion-weighted neighbours               | gene-specific | predictor-dependent       | neighbourhood | to be assessed                              | structural                                                         |
| Multi-network integration (GeneMANIA)                                    | (network, neighbour) pairs with weights     | gene-specific | predictor-dependent       | neighbourhood | to be assessed                              | structural                                                         |
| Sequence-homology hit lists                                              | homologues with alignment scores            | gene-specific | predictor-dependent       | neighbourhood | to be assessed                              | structural                                                         |
| Embedding $k$ -nearest neighbours                                        | nearest neighbours with distances           | gene-specific | predictor-dependent       | neighbourhood | to be assessed                              | structural                                                         |

**Table S8.** Neighbourhood-structure family: gain over the exhaustive matched-risk frontier of the global path, over 100 splits. For each arm and split, the frontier is the best yield among all global score cutoffs, including abstention, whose realised held-out risk does not exceed that arm's realised risk.

| Data   | $\alpha$ | $\delta$ | GoCo-M                     |              | GoCo-N                     |              |
|--------|----------|----------|----------------------------|--------------|----------------------------|--------------|
|        |          |          | Correct calls vs. frontier | Splits ahead | Correct calls vs. frontier | Splits ahead |
| FunMap | 0.05     | 0.10     | +101                       | 1.00         | +155                       | 0.95         |
|        | 0.05     | 0.50     | +99                        | 1.00         | +175                       | 0.97         |
|        | 0.10     | 0.10     | +196                       | 1.00         | +379                       | 1.00         |
|        | 0.10     | 0.50     | +190                       | 1.00         | +393                       | 1.00         |
|        | 0.20     | 0.10     | +164                       | 1.00         | +519                       | 1.00         |
|        | 0.20     | 0.50     | +186                       | 1.00         | +535                       | 1.00         |
| STRING | 0.05     | 0.10     | +46                        | 1.00         | +172                       | 1.00         |
|        | 0.05     | 0.50     | +40                        | 1.00         | +186                       | 1.00         |
|        | 0.10     | 0.10     | +138                       | 1.00         | +268                       | 0.97         |
|        | 0.10     | 0.50     | +135                       | 1.00         | +282                       | 0.98         |
|        | 0.20     | 0.10     | +197                       | 1.00         | +561                       | 0.99         |
|        | 0.20     | 0.50     | +205                       | 1.00         | +599                       | 0.99         |

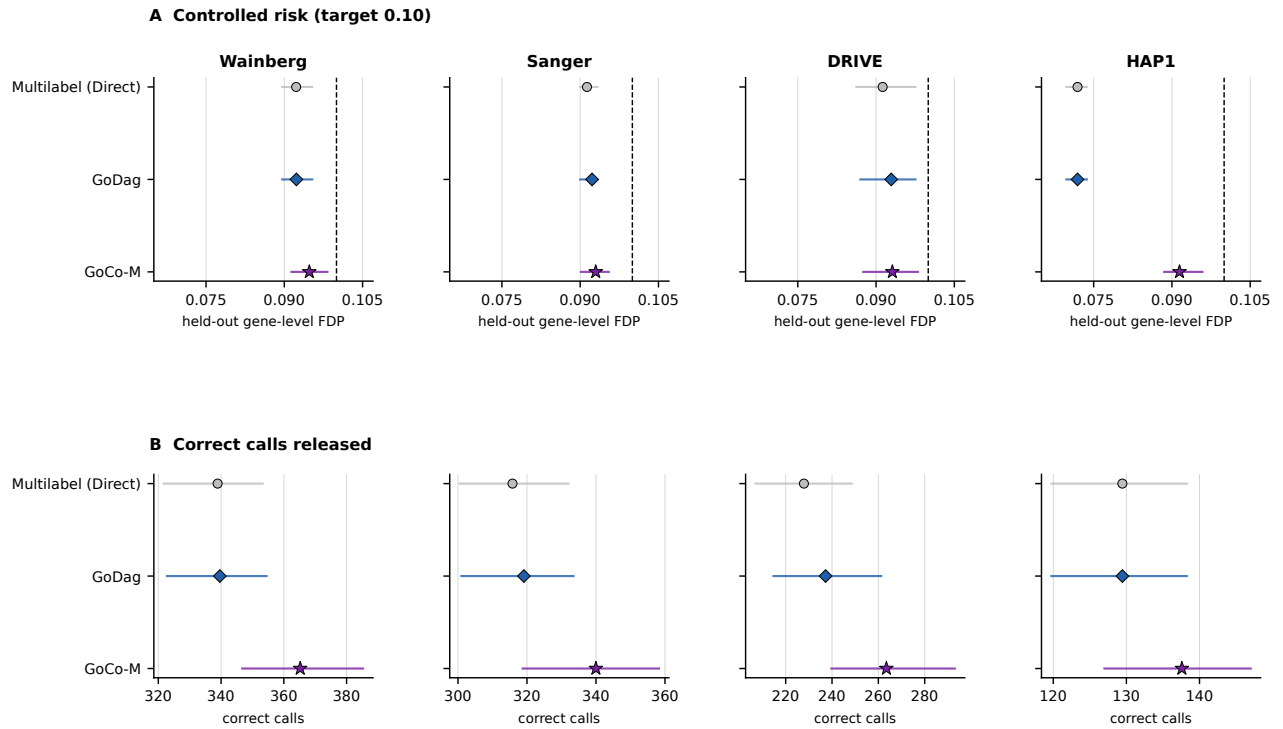
**Table S9.** Ranking-fold selection between GoCo-M and GoCo-N over 100 splits at  $\delta = 0.10$ . In each split, the rule selects the instantiation with larger ranking-fold supported yield at the last candidate whose ranking-fold mean loss does not exceed  $\alpha$ .

| Predictor | Evidence      | $\alpha$ | $\mathcal{T}$ yield |        | Rule chooses<br>GoCo-N | Held-out supported |        |          | Pr(N > M)<br>held-out | Rule agrees<br>with winner |
|-----------|---------------|----------|---------------------|--------|------------------------|--------------------|--------|----------|-----------------------|----------------------------|
|           |               |          | GoCo-M              | GoCo-N |                        | GoCo-M             | GoCo-N | selected |                       |                            |
| Wainberg  | module-shared | 0.05     | 68.5                | 55.7   | 0.16                   | 213.2              | 168.5  | 206.1    | 0.07                  | 0.81                       |
|           |               | 0.10     | 117.7               | 73.3   | 0.00                   | 365.2              | 222.8  | 365.2    | 0.00                  | 1.00                       |
|           |               | 0.20     | 176.7               | 131.2  | 0.00                   | 539.2              | 392.8  | 539.2    | 0.00                  | 1.00                       |
| Sanger    | module-shared | 0.05     | 77.5                | 114.3  | 0.85                   | 213.7              | 351.5  | 331.3    | 0.99                  | 0.84                       |
|           |               | 0.10     | 127.7               | 185.3  | 0.96                   | 340.0              | 559.2  | 553.4    | 1.00                  | 0.96                       |
|           |               | 0.20     | 478.9               | 495.1  | 0.61                   | 1346.8             | 1430.8 | 1411.7   | 0.78                  | 0.65                       |
| DRIVE     | module-shared | 0.05     | 36.3                | 37.8   | 0.52                   | 94.2               | 97.4   | 96.8     | 0.52                  | 0.54                       |
|           |               | 0.10     | 98.0                | 72.0   | 0.18                   | 263.5              | 194.5  | 251.5    | 0.02                  | 0.80                       |
|           |               | 0.20     | 231.1               | 225.9  | 0.00                   | 702.0              | 701.5  | 702.0    | 0.00                  | 1.00                       |
| HAP1      | module-shared | 0.05     | 32.9                | 49.3   | 0.86                   | 102.4              | 163.3  | 156.9    | 0.97                  | 0.83                       |
|           |               | 0.10     | 44.6                | 67.6   | 0.89                   | 137.6              | 225.1  | 217.5    | 1.00                  | 0.89                       |
|           |               | 0.20     | 61.8                | 90.6   | 0.90                   | 187.7              | 299.3  | 290.2    | 1.00                  | 0.90                       |
| FunMap    | neighbourhood | 0.05     | 319.4               | 349.8  | 0.73                   | 954.5              | 1016.1 | 996.7    | 0.72                  | 0.61                       |
|           |               | 0.10     | 488.3               | 552.9  | 0.92                   | 1474.4             | 1615.8 | 1608.1   | 0.89                  | 0.83                       |
|           |               | 0.20     | 759.2               | 885.5  | 0.96                   | 2157.9             | 2615.8 | 2598.2   | 1.00                  | 0.96                       |
| STRING    | neighbourhood | 0.05     | 172.1               | 221.5  | 0.89                   | 503.6              | 621.4  | 610.0    | 0.96                  | 0.86                       |
|           |               | 0.10     | 350.2               | 395.6  | 0.76                   | 1021.1             | 1151.1 | 1124.5   | 0.89                  | 0.69                       |
|           |               | 0.20     | 642.1               | 766.6  | 0.93                   | 1732.5             | 2243.0 | 2201.7   | 0.99                  | 0.92                       |

**Table S10.** Comparison with validation  $F_{\max}$  and top- $k$  release rules on the same 100 split streams and held-out folds. The  $F_{\max}$  threshold is fitted on the 70% calibration sample using the CAFA protein-centric  $F_1$  measure; GoCo is shown at  $\alpha = 0.10$ ,  $\delta = 0.10$ .

| Predictor | Release rule                                                       | Threshold | Validation $F$ | Gene FDP     | Pr( $\hat{R}_{\mathcal{E}} > 0.10$ ) | Genes with a call | Calls  | Correct calls |
|-----------|--------------------------------------------------------------------|-----------|----------------|--------------|--------------------------------------|-------------------|--------|---------------|
| Wainberg  | Validation $F_{\max}$ threshold                                    | 602       | 0.002          | 0.231        | 1.00                                 | 1232              | 4941   | 559           |
|           | Top-1 GO term per gene                                             | –         | –              | 0.223        | 1.00                                 | 1232              | 1232   | 175           |
|           | Top-3 GO terms per gene                                            | –         | –              | 0.228        | 1.00                                 | 1232              | 2759   | 370           |
|           | <b>GoCo, certified at <math>\alpha = 0.10</math></b> nested policy | –         | –              | <b>0.095</b> | <b>0.13</b>                          | 568               | 1896   | <b>365</b>    |
| Sanger    | Validation $F_{\max}$ threshold                                    | 100       | 0.006          | 0.224        | 1.00                                 | 1094              | 21840  | 1685          |
|           | Top-1 GO term per gene                                             | –         | –              | 0.211        | 1.00                                 | 1094              | 1094   | 150           |
|           | Top-3 GO terms per gene                                            | –         | –              | 0.214        | 1.00                                 | 1094              | 3119   | 401           |
|           | <b>GoCo, certified at <math>\alpha = 0.10</math></b> nested policy | –         | –              | <b>0.093</b> | <b>0.15</b>                          | 505               | 1983   | <b>340</b>    |
| DRIVE     | Validation $F_{\max}$ threshold                                    | 100       | 0.004          | 0.184        | 1.00                                 | 378               | 8370   | 702           |
|           | Top-1 GO term per gene                                             | –         | –              | 0.173        | 1.00                                 | 378               | 378    | 51            |
|           | Top-3 GO terms per gene                                            | –         | –              | 0.175        | 1.00                                 | 378               | 1044   | 141           |
|           | <b>GoCo, certified at <math>\alpha = 0.10</math></b> nested policy | –         | –              | <b>0.093</b> | <b>0.16</b>                          | 208               | 1987   | <b>263</b>    |
| HAP1      | Validation $F_{\max}$ threshold                                    | 100       | 0.004          | 0.986        | 1.00                                 | 4506              | 102090 | 1341          |
|           | Top-1 GO term per gene                                             | –         | –              | 0.979        | 1.00                                 | 4506              | 4506   | 94            |
|           | Top-3 GO terms per gene                                            | –         | –              | 0.982        | 1.00                                 | 4506              | 13508  | 240           |
|           | <b>GoCo, certified at <math>\alpha = 0.10</math></b> nested policy | –         | –              | <b>0.091</b> | <b>0.06</b>                          | 462               | 1252   | <b>138</b>    |
| FunMap    | Validation $F_{\max}$ threshold                                    | 1         | 0.069          | 0.787        | 1.00                                 | 2912              | 80278  | 11828         |
|           | Top-1 GO term per gene                                             | –         | –              | 0.599        | 1.00                                 | 2912              | 2912   | 1057          |
|           | Top-3 GO terms per gene                                            | –         | –              | 0.642        | 1.00                                 | 2912              | 8526   | 2746          |
|           | <b>GoCo, certified at <math>\alpha = 0.10</math></b> nested policy | –         | –              | <b>0.088</b> | <b>0.05</b>                          | 1084              | 2166   | <b>1616</b>   |
| STRING    | Validation $F_{\max}$ threshold                                    | 1         | 0.098          | 0.828        | 1.00                                 | 2375              | 156273 | 19717         |
|           | Top-1 GO term per gene                                             | –         | –              | 0.591        | 1.00                                 | 2375              | 2375   | 946           |
|           | Top-3 GO terms per gene                                            | –         | –              | 0.635        | 1.00                                 | 2375              | 7118   | 2517          |
|           | <b>GoCo, certified at <math>\alpha = 0.10</math></b> nested policy | –         | –              | <b>0.089</b> | <b>0.06</b>                          | 806               | 1622   | <b>1151</b>   |

$\alpha = 0.10, \delta = 0.10$ : mean over 100 splits (marker) and interquartile range (bar)



**Fig. S1.** Primary comparison of Multilabel, GoDag and GoCo-M at  $\alpha = 0.10, \delta = 0.10$ : held-out gene-level TruePath FDP and reference-supported calls over 100 splits.



**Table S11.** Prespecified constants of GoCo. Module-family choices were fixed on Wainberg and the neighbourhood-family instantiation on FunMap; none uses certification- or evaluation-fold labels, and none affects validity (main-text Proposition 1).

| Constant                                           | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Split                                              | 10% $\mathcal{T}$ / 60% $\mathcal{C}$ / 30% $\mathcal{E}$ by seeded permutation (20260905 + $s$ in the module family and 20260910 + $s$ in the neighbourhood family, $s = 0, \dots, 99$ ); the instantiations were fixed on streams 0–9 of Wainberg and FunMap, respectively; $n_{\mathcal{T}} = \text{round}(0.10n)$ and $n_{\mathcal{T}} + m = \text{round}(0.70n)$ with round-half-to-even; module-family global methods calibrate on $\mathcal{T} \cup \mathcal{C}$ (70%), whereas all neighbourhood-family methods certify on $\mathcal{C}$ (60%) |
| GoCo-M global grid $\lambda_1 > \dots > \lambda_J$ | 3000, 2500, 2000, 1600, 1400, 1200, 1100, 1050, then 1000 downward in steps of 25 (to 600 in Wainberg, $J = 25$ ; to 100 in Sanger, DRIVE and HAP1, $J = 45$ ); identical for GoDag and GoCo-M in the module family                                                                                                                                                                                                                                                                                                                                    |
| GoCo-M block insertion trigger                     | ranking-fold mean loss at $\lambda_{j+1}$ exceeds $0.8\alpha$ (i.e. 0.08 at $\alpha = 0.10$ ) and $ \mathcal{T}_j  \geq 5$                                                                                                                                                                                                                                                                                                                                                                                                                             |
| GoCo-M admission grid $\mathcal{Q}$                | 0.01, 0.02, 0.03, 0.04, 0.05, 0.075, 0.10, 0.125, 0.15, 0.175, 0.20, 0.225, 0.25, 0.275, 0.30, 0.35, 0.40, 0.50, 0.60, 0.75 ( $q = 1$ is the next global threshold)                                                                                                                                                                                                                                                                                                                                                                                    |
| Shrinkage weight                                   | $\kappa = 1$ in Equation (S3); two pseudo-calls for the per-call rate features                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Ranking statistic                                  | GoCo-M: Equation (S4); GoCo-N: individual calls ordered by $\hat{p}_{ig}$ , with smaller predicted unsupported probability entering first; both use the cross-fitted logistic per-call model ( $C = 0.5$ , standardised features, two hash-defined halves of $\mathcal{T}$ )                                                                                                                                                                                                                                                                           |
| Instantiation                                      | GoCo-M (partial admission within grid steps) for evidence shared between genes, fixed on Wainberg; GoCo-N (nested top- $\ell$ calls by $\hat{p}_{ig}$ ) for per-gene neighbourhood evidence, fixed on FunMap                                                                                                                                                                                                                                                                                                                                           |
| Tie-break                                          | GoCo-M: deterministic lexicographic ordering by $(u_i, h_i, \text{gene identifier})$ , where $h_i$ is the SHA-256-derived hash of the gene identifier; GoCo-N: ties in $\hat{p}_{ig}$ are broken by the frozen gene hash and then by call index; exact comparisons, no numerical tolerance                                                                                                                                                                                                                                                             |
| Certification                                      | $\alpha = 0.10$ (sensitivity 0.05, 0.20); $\delta = 0.10$ and $\delta = 0.50$ ; $\hat{\sigma}_k$ floored at $10^{-6}$                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| GO reference                                       | Direct annotations and TruePath closure frozen at one release for all four dependency data sets (GO release 2019-11-01: <code>goa.human.gaf</code> and <code>go-basic.obo</code> from the GO archive; all evidence codes; NOT-qualified annotations and root terms removed; closure over <code>is_a</code> and <code>part_of</code> ); GO archive 2024-09-08 for FunMap and STRING                                                                                                                                                                     |

**Table S12.** Methods compared. All are post-prediction release rules applied to the same frozen scores and split streams; columns distinguish the certified loss, candidate path, calibration sample and use of GO hierarchy and predictor-native evidence.

| Method                           | Loss certified        | Candidate path                                                                               | Calibration sample                                               | GO hierarchy | Predictor-native evidence             |
|----------------------------------|-----------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------|---------------------------------------|
| $F_{\max}$ threshold             | none                  | global cutoff maximising $F_1$                                                               | (threshold fitted on 70%)                                        | no           | no                                    |
| Top- $k$ per gene                | none                  | the $k$ highest-scoring terms of each gene                                                   | none                                                             | no           | no                                    |
| Multilabel [24]                  | Direct FDP            | global grid $\lambda_1 > \dots > \lambda_J$                                                  | 70% (module family); $\mathcal{C}$ (60%) in neighbourhood family | no           | no                                    |
| Generic calibrators (nine rules) | Direct FDP            | global grid                                                                                  | 70% (module family); $\mathcal{C}$ (60%) in neighbourhood family | no           | no                                    |
| GoDag                            | TruePath FDP, Eq. (1) | global grid                                                                                  | 70% (module family); $\mathcal{C}$ (60%) in neighbourhood family | yes          | no                                    |
| GoCo-M                           | TruePath FDP          | grid, plus partial admission of affected genes within each step, ordered by $u_i$ (Eq. (S4)) | $\mathcal{C}$ (60%); ordering fitted on $\mathcal{T}$ (10%)      | yes          | yes, at gene resolution within a step |
| GoCo-N                           | TruePath FDP          | top- $\ell$ calls ordered by $\hat{p}_{ig}$                                                  | $\mathcal{C}$ ; model fitted on $\mathcal{T}$                    | yes          | yes, at call resolution               |

**Table S13.** Error-target sweep at  $\delta = 0.50$ : mean held-out gene-level FDP and supported GO-term yield at  $\alpha \in \{0.05, 0.10, 0.20\}$  for every predictor, 100 splits.

| Data set                                                     | Method                                | $\alpha = 0.05$ |               | $\alpha = 0.10$ |               | $\alpha = 0.20$ |               |
|--------------------------------------------------------------|---------------------------------------|-----------------|---------------|-----------------|---------------|-----------------|---------------|
|                                                              |                                       | FDP             | Correct calls | FDP             | Correct calls | FDP             | Correct calls |
| <i>Module structure: GoCo-M (development: Wainberg)</i>      |                                       |                 |               |                 |               |                 |               |
| Wainberg                                                     | Multilabel (Direct loss)              | 0.0359          | 183.1         | 0.0923          | 339.6         | 0.1699          | 518.9         |
|                                                              | GoDag                                 | 0.0359          | 183.1         | 0.0923          | 339.6         | 0.1699          | 518.9         |
|                                                              | <b>GoCo-M</b>                         | 0.0466          | 214.8         | 0.0964          | 369.1         | 0.1939          | 539.6         |
| Sanger                                                       | Multilabel (Direct loss)              | 0.0474          | 212.1         | 0.0953          | 333.0         | 0.1895          | 1163.2        |
|                                                              | GoDag                                 | 0.0484          | 218.2         | 0.0980          | 353.8         | 0.1895          | 1163.2        |
|                                                              | <b>GoCo-M</b>                         | 0.0491          | 234.5         | 0.0980          | 380.7         | 0.1913          | 1360.9        |
| DRIVE                                                        | Multilabel (Direct loss)              | 0.0444          | 92.3          | 0.0945          | 248.1         | 0.1844          | 702.4         |
|                                                              | GoDag                                 | 0.0466          | 94.4          | 0.0965          | 258.6         | 0.1844          | 702.4         |
|                                                              | <b>GoCo-M</b>                         | 0.0482          | 99.4          | 0.0971          | 291.7         | 0.1844          | 702.4         |
| HAP1                                                         | Multilabel (Direct loss)              | 0.0436          | 96.1          | 0.0719          | 129.4         | 0.1353          | 160.2         |
|                                                              | GoDag                                 | 0.0436          | 96.1          | 0.0719          | 129.4         | 0.1353          | 160.2         |
|                                                              | <b>GoCo-M</b>                         | 0.0459          | 102.7         | 0.0943          | 137.7         | 0.1872          | 188.3         |
| <i>Neighbourhood structure: GoCo-N (development: FunMap)</i> |                                       |                 |               |                 |               |                 |               |
| FunMap                                                       | Multilabel (Direct loss, shared grid) | 0.0312          | 727.9         | 0.0741          | 1129.7        | 0.1543          | 1869.5        |
|                                                              | GoDag, shared grid                    | 0.0312          | 727.9         | 0.0901          | 1261.2        | 0.1722          | 1995.4        |
|                                                              | GoDag, refined grid                   | 0.0486          | 911.0         | 0.0986          | 1324.0        | 0.1968          | 2189.4        |
|                                                              | <b>GoCo-N</b>                         | 0.0462          | 1065.8        | 0.0921          | 1665.8        | 0.1881          | 2671.8        |
| STRING                                                       | Multilabel (Direct loss, shared grid) | 0.0386          | 424.9         | 0.0828          | 828.1         | 0.1458          | 1383.8        |
|                                                              | GoDag, shared grid                    | 0.0386          | 424.9         | 0.0828          | 828.1         | 0.1693          | 1562.5        |
|                                                              | GoDag, refined grid                   | 0.0487          | 496.1         | 0.0978          | 962.1         | 0.1967          | 1768.4        |
|                                                              | <b>GoCo-N</b>                         | 0.0466          | 663.9         | 0.0944          | 1208.0        | 0.1905          | 2332.3        |

FDP: mean held-out gene-level TruePath false discovery proportion. Correct calls: released gene-GO pairs the frozen reference confirms. Each structure is shown with the instantiation prespecified for it. Module-family GoCo-M and GoDag share the 25-unit grid. In the neighbourhood family, Multilabel and GoDag are shown on the shared coarse grid, with refined-grid GoDag separately; GoCo-N uses a distinct call-ordered path. Grid refinement does not change the fixed-sequence multiplicity argument.